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(54) **SYSTEM AND METHOD FOR PROVIDING
REAL-TIME SPEECH RECOGNITION AND
NATIVE VISUALIZATION FOR DATA
ANALYTICS**

(71) Applicant: **ORACLE INTERNATIONAL
CORPORATION**, Redwood Shores,
CA (US)

(72) Inventors: **Sergio Alejandro Acosta Moreno**,
Zapopan (MX); **Rodolfo Javier Melin
Ruiz**, Tlajomulco de Zúñiga (MX);
Mathew Milella, St. Charles, IL (US);
Jacques Vigeant, Fort Lauderdale, FL
(US)

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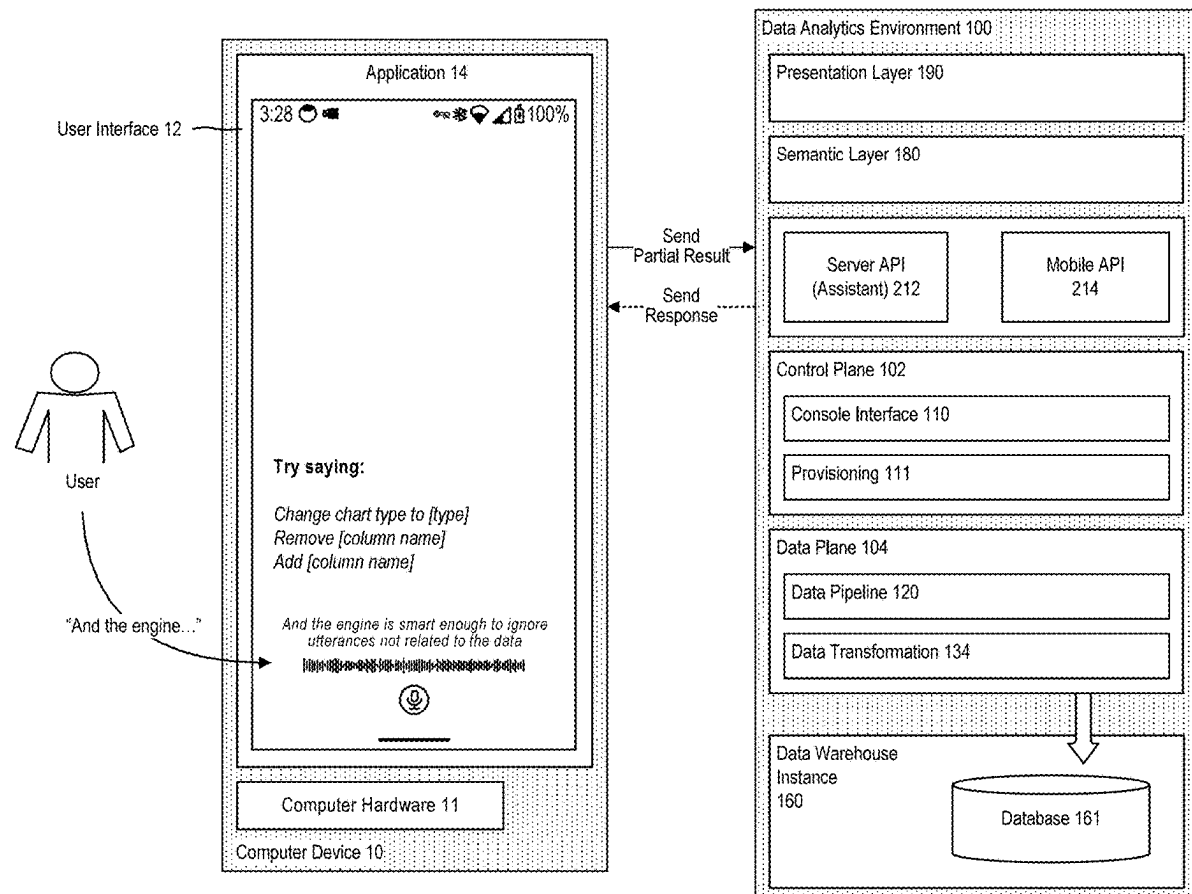
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G10L 15/30 (2013.01)
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(57) ABSTRACT

Embodiments described herein are generally related to computer data analytics, and computer-based methods of providing business intelligence or other data, and are particularly related to a system and method for providing a real-time speech recognition and native visualization tool, for use with data analytics environments. The described approach can be used to provide real-time data visualizations or other analytics based on a live conversation with a user using an on-device speech-to-text recognition engine. In accordance with an embodiment, the method includes: listening for audio received as input from a user; building an array of words associated with a sentence; determining whether the sentence is capable of being processed without additional server data and if so performing the step locally; and alternatively sending an input sentence to a server API, in order to determine an intent associated therewith for use by the server in returning relevant results.



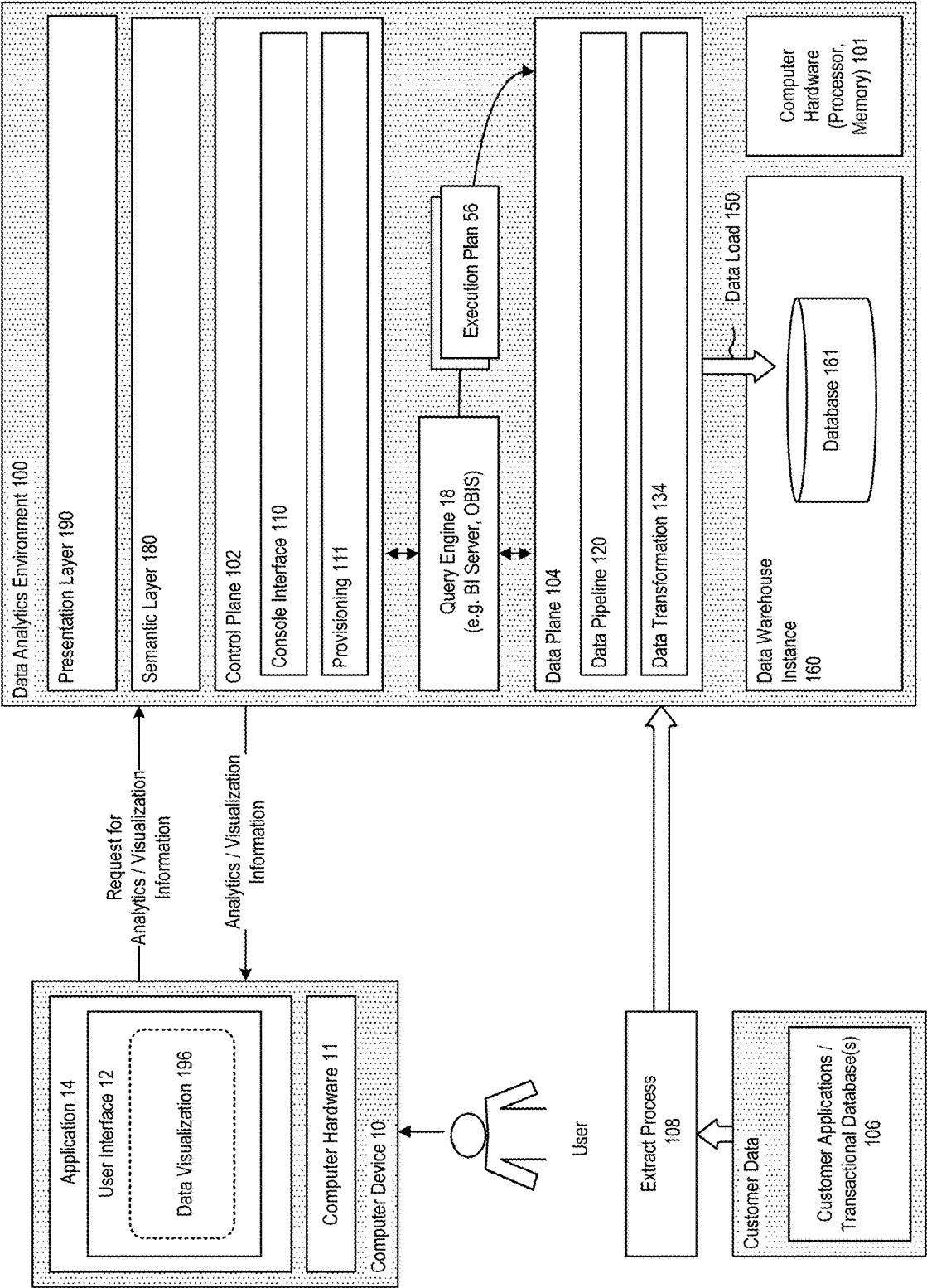


FIGURE 1

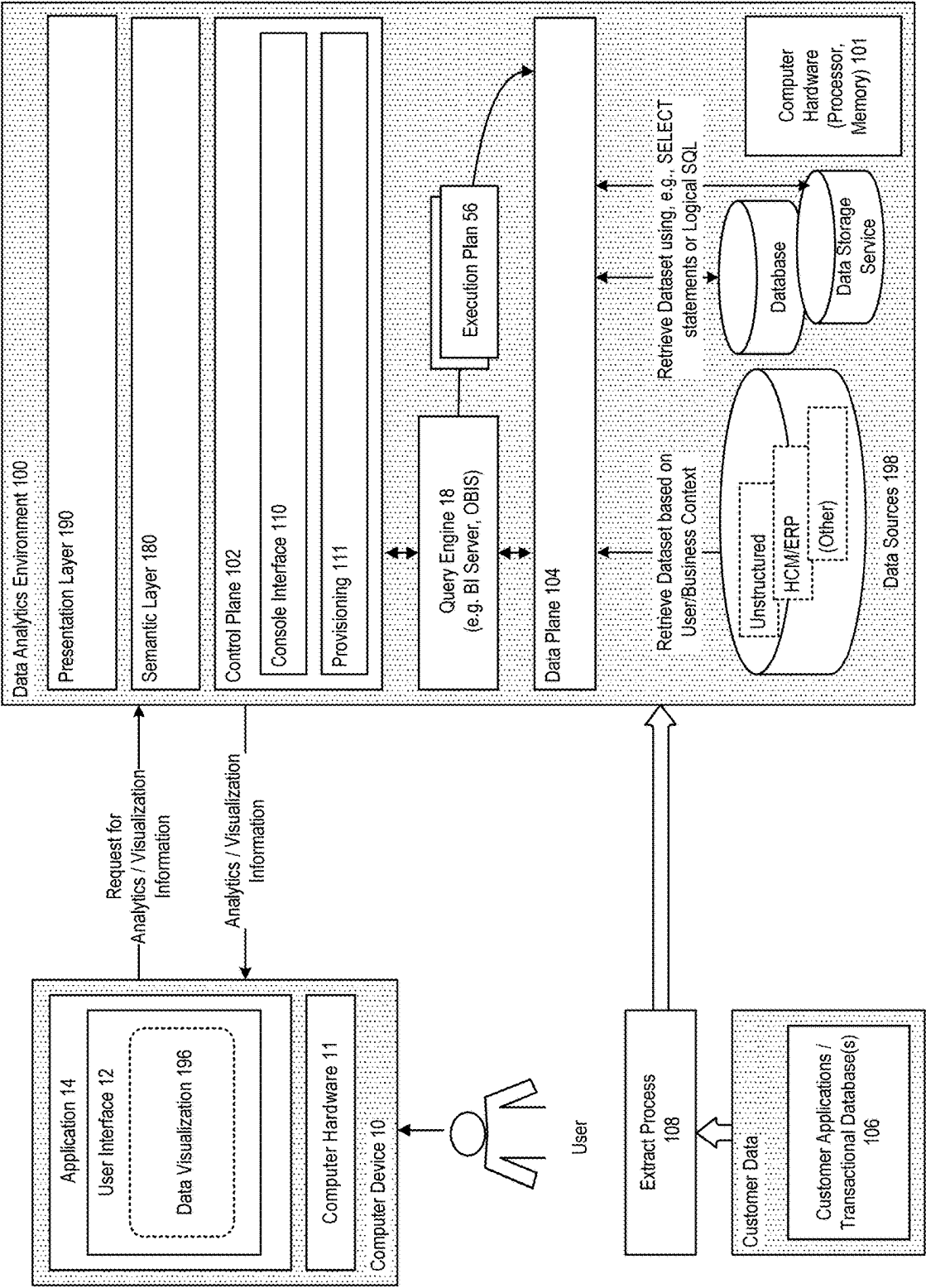


FIGURE 2

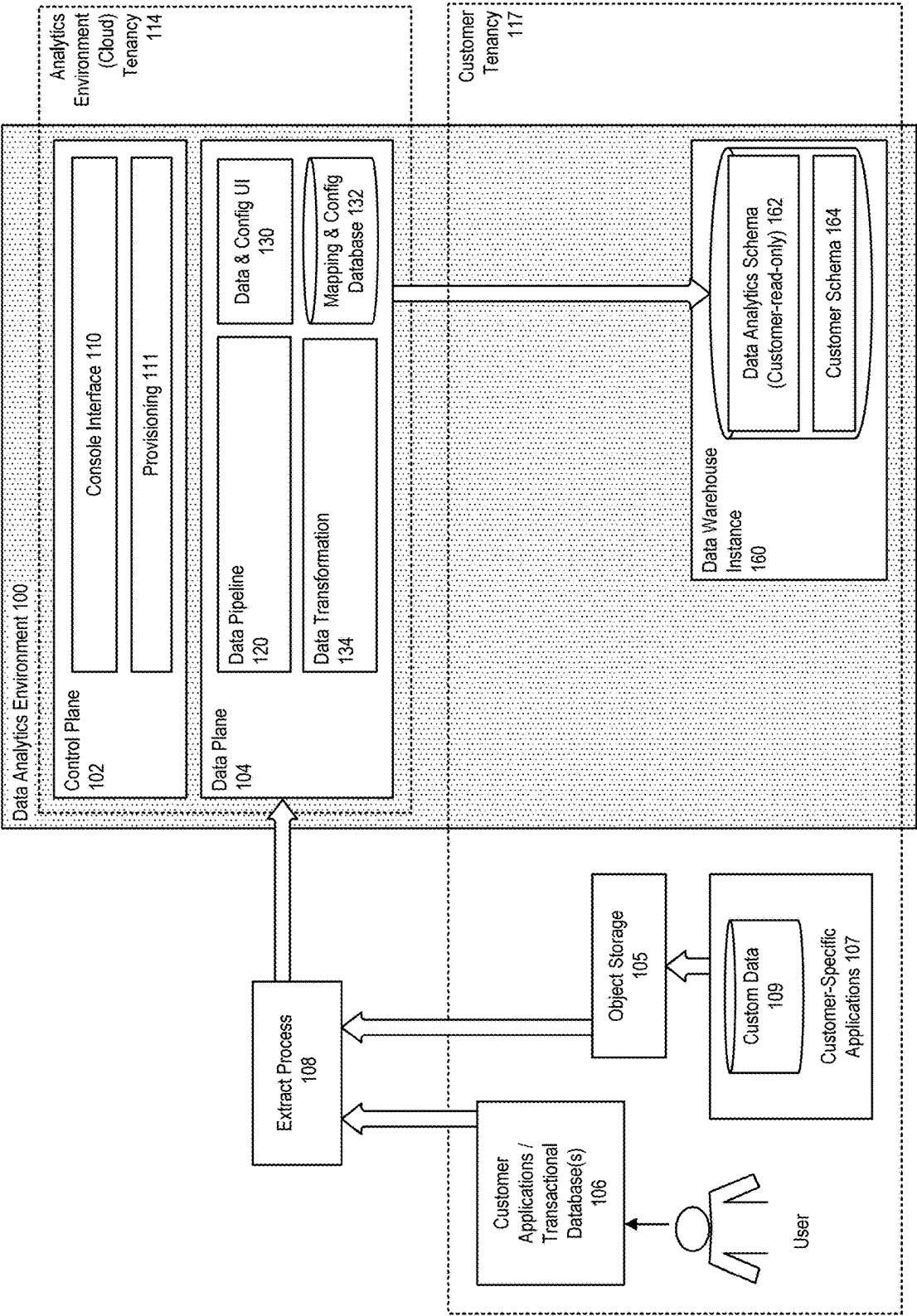


FIGURE 3

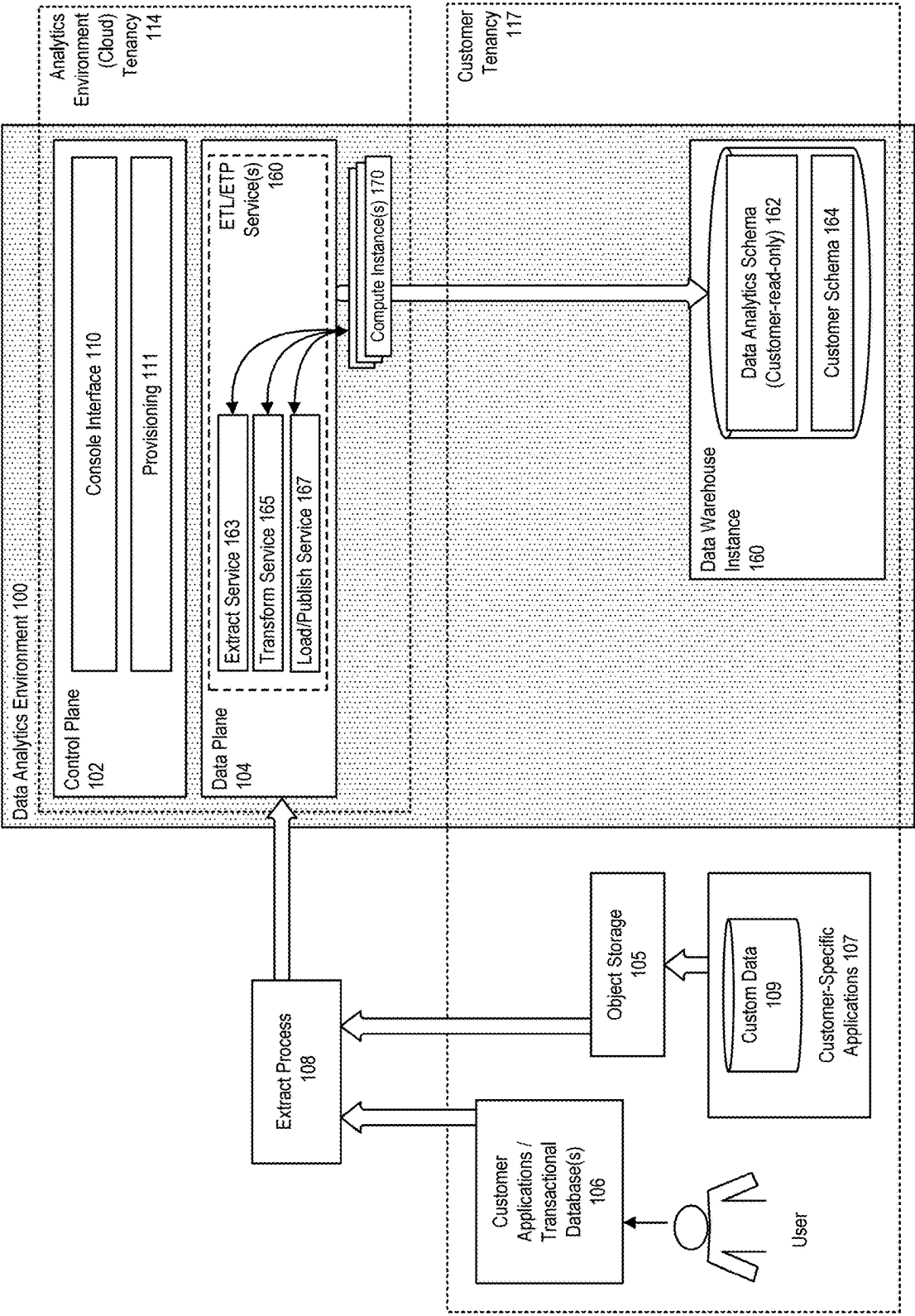


FIGURE 4

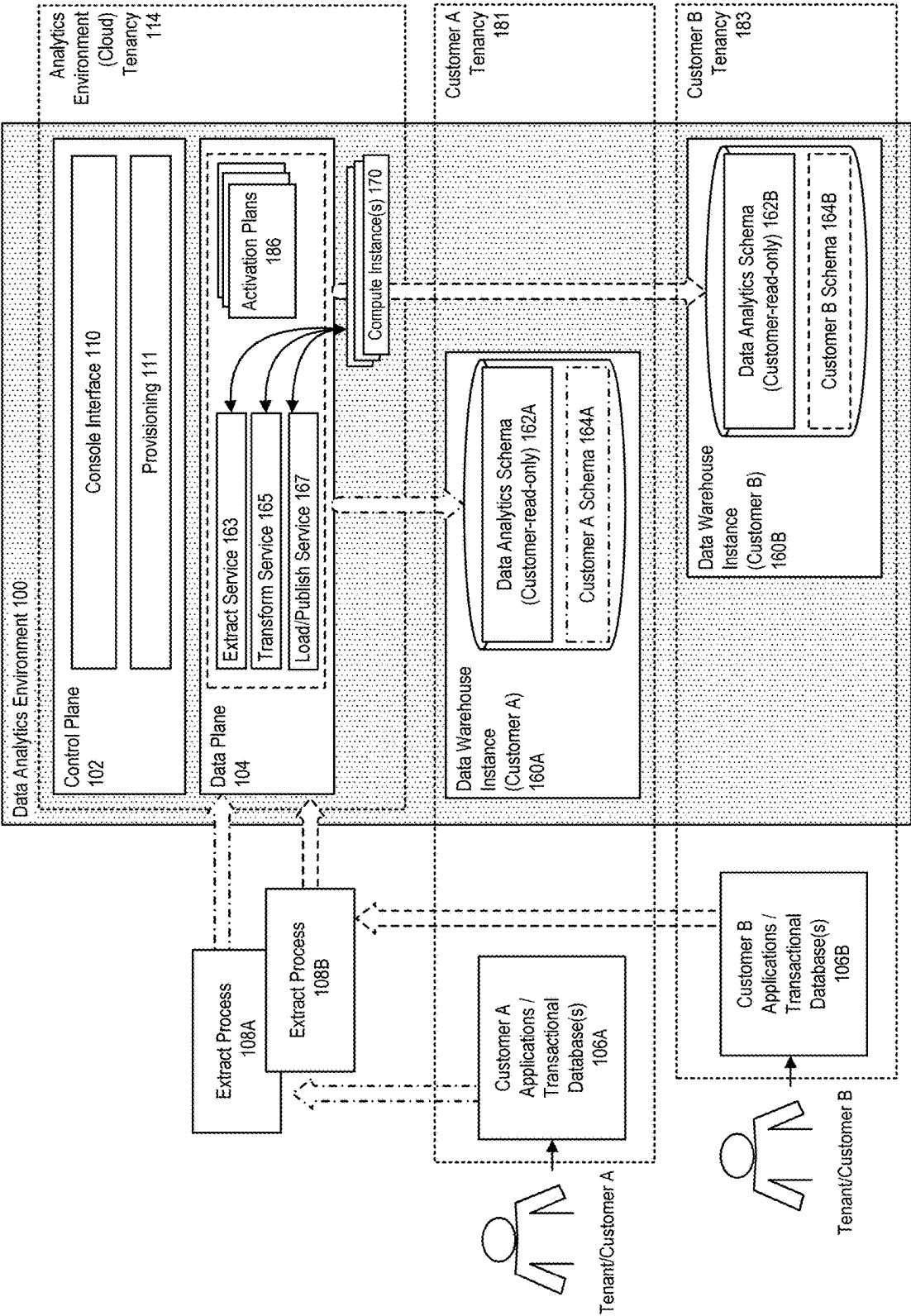


FIGURE 5

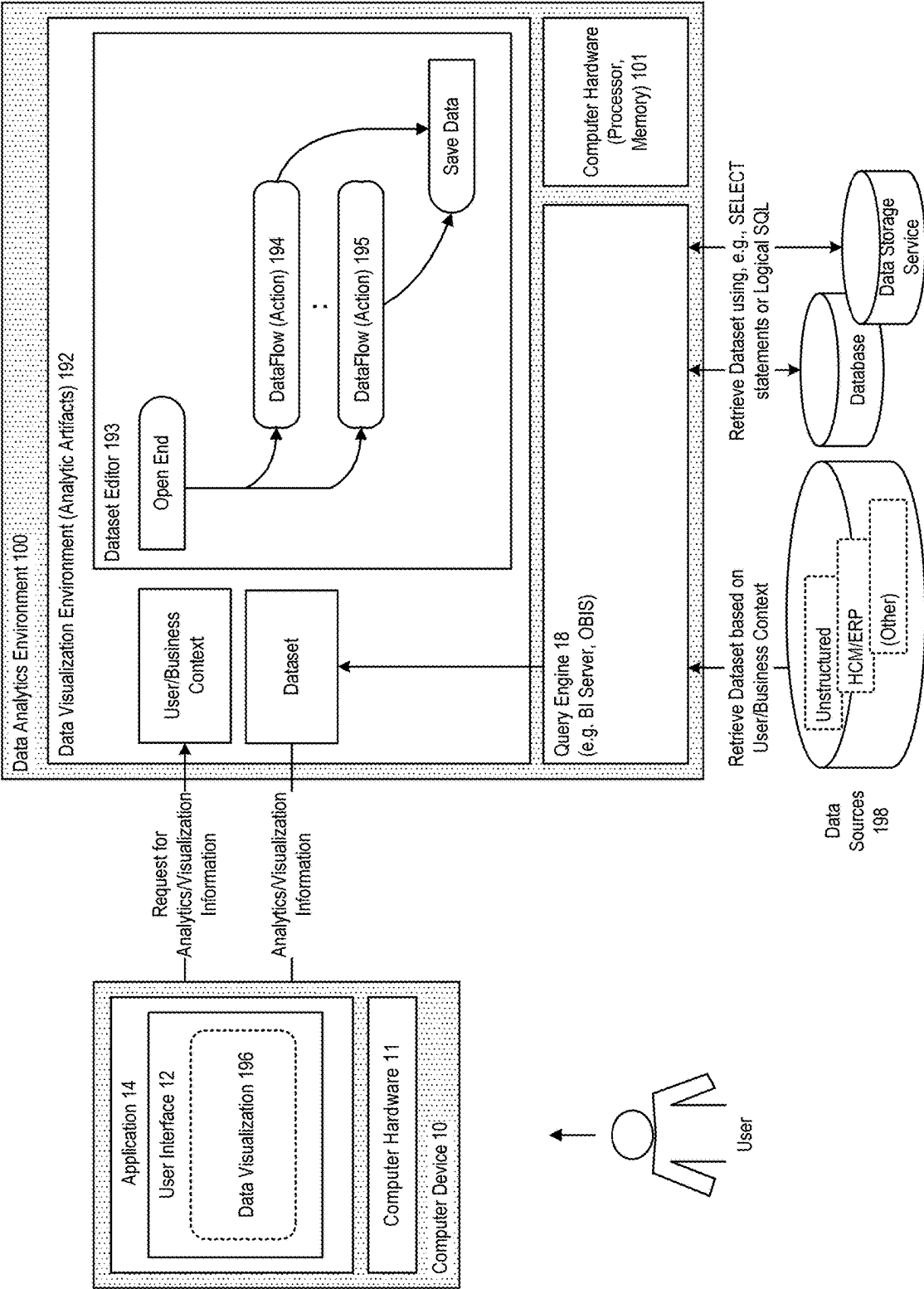


FIGURE 6

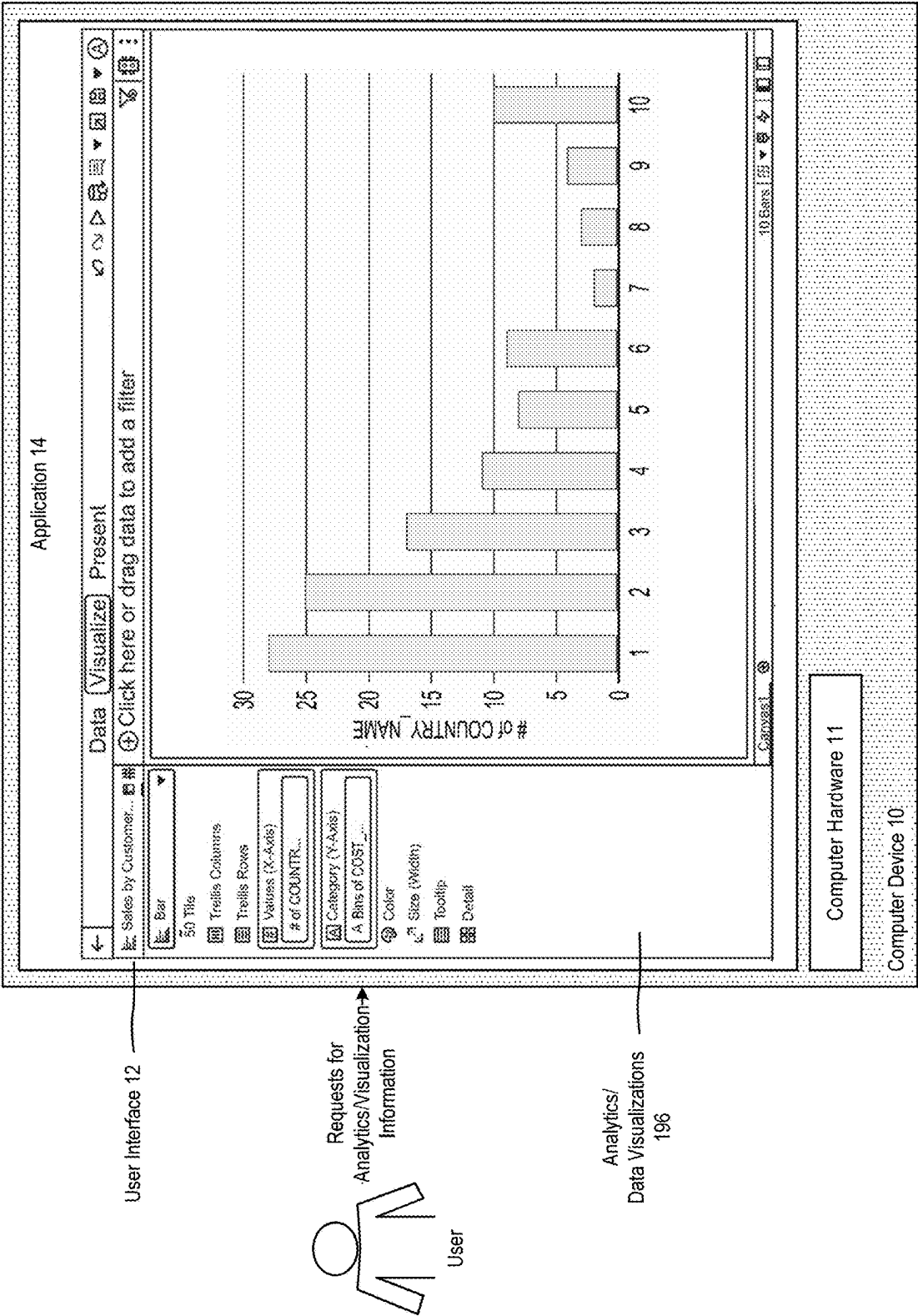


FIGURE 7

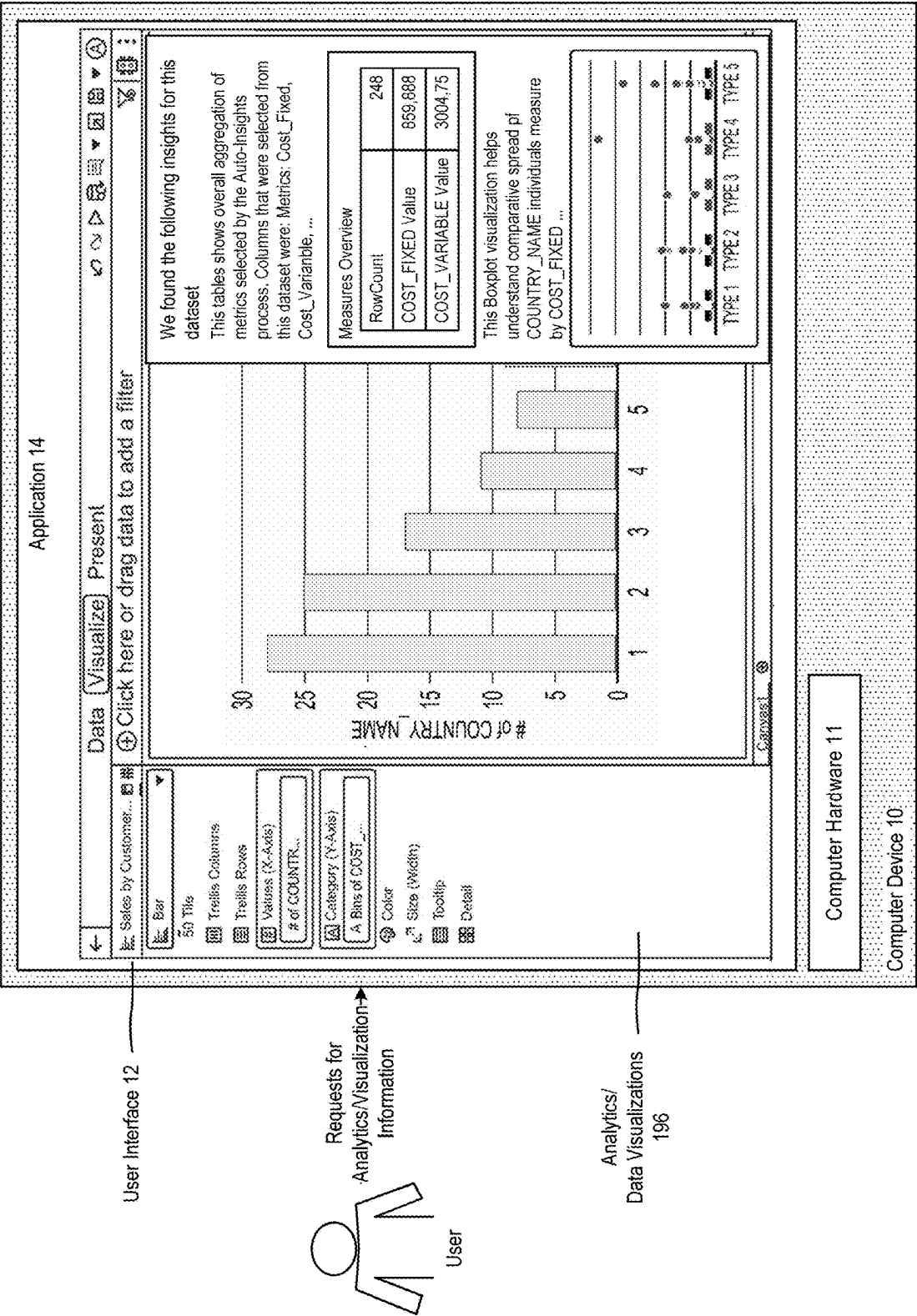


FIGURE 8

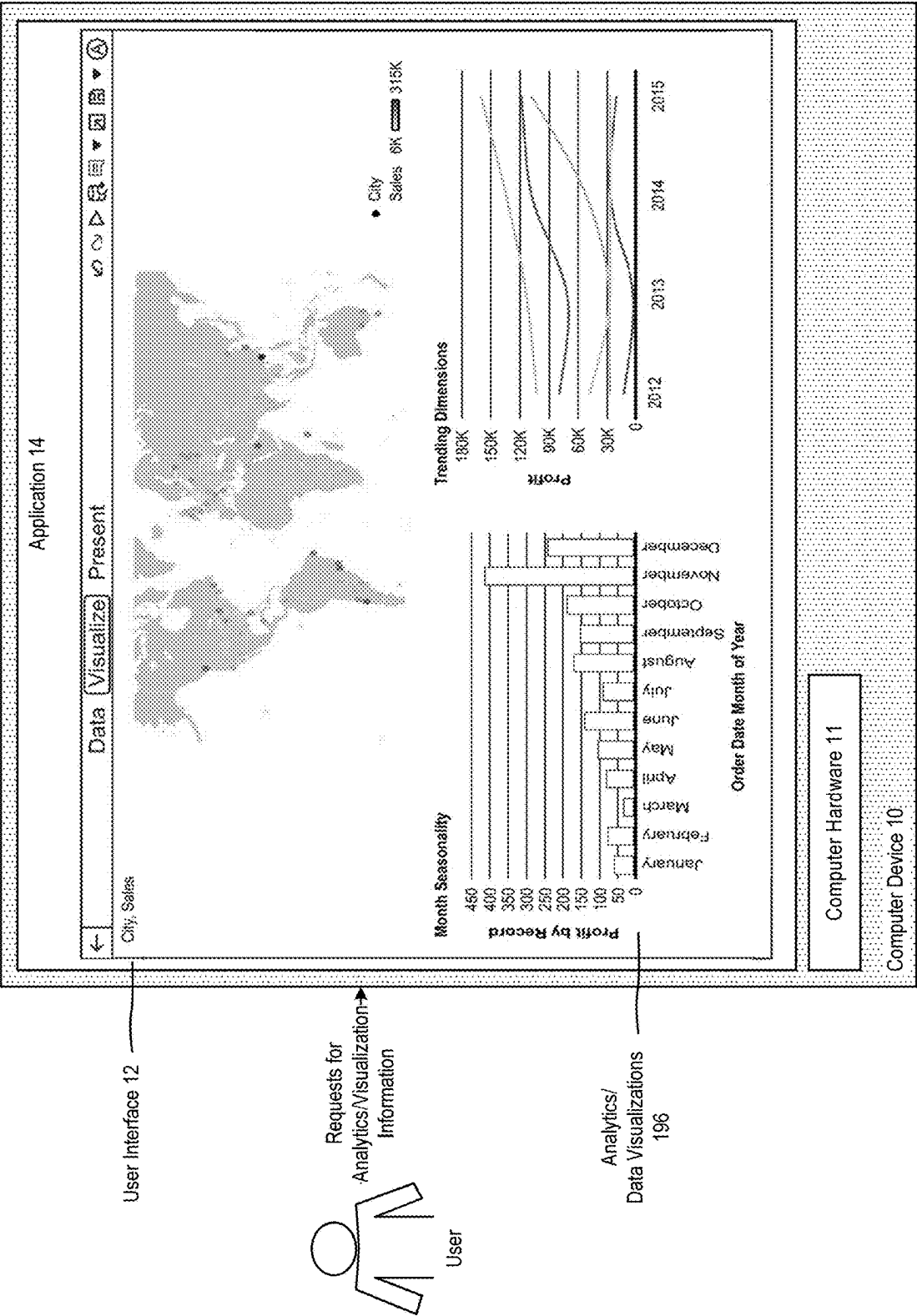


FIGURE 9

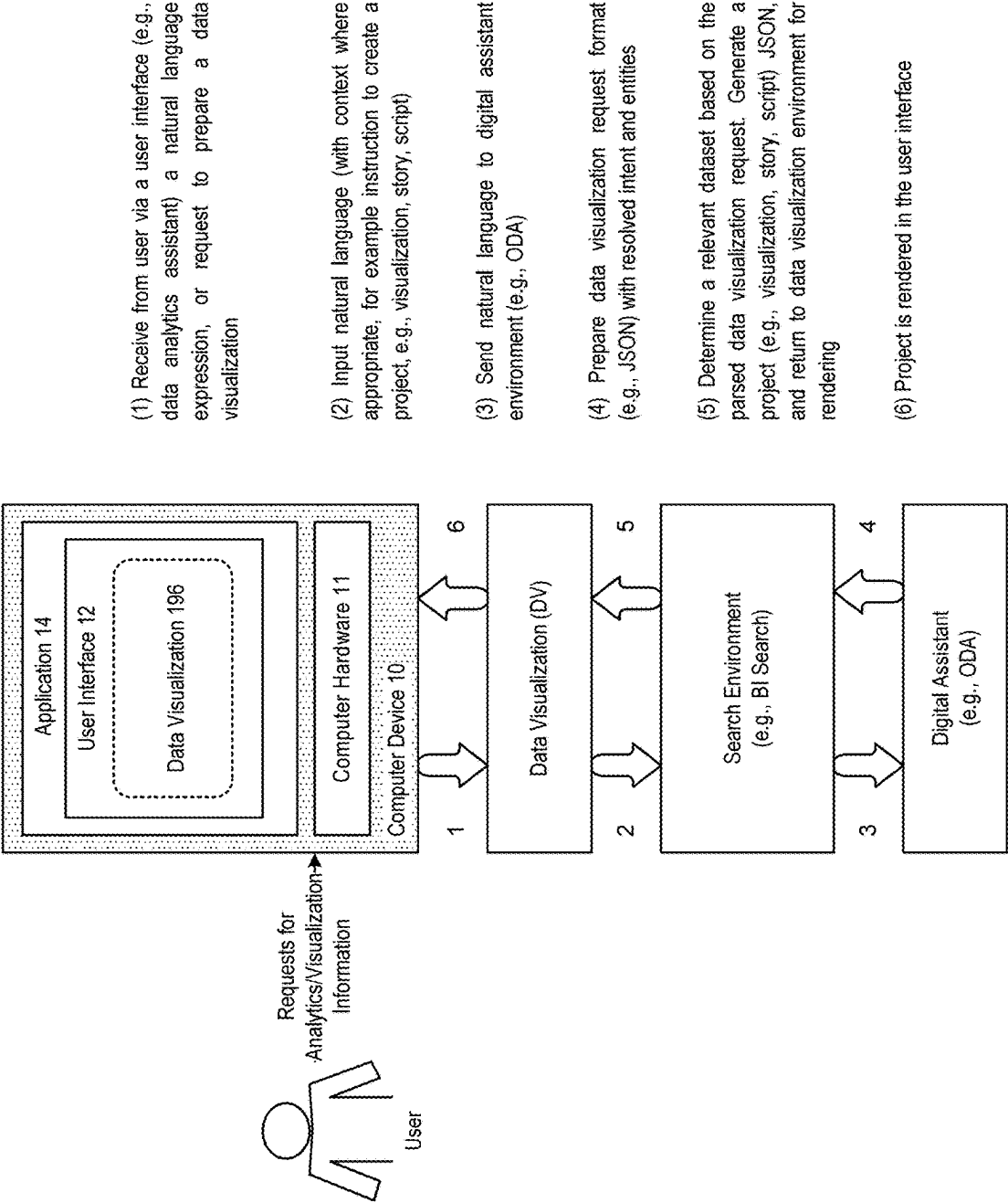


FIGURE 10

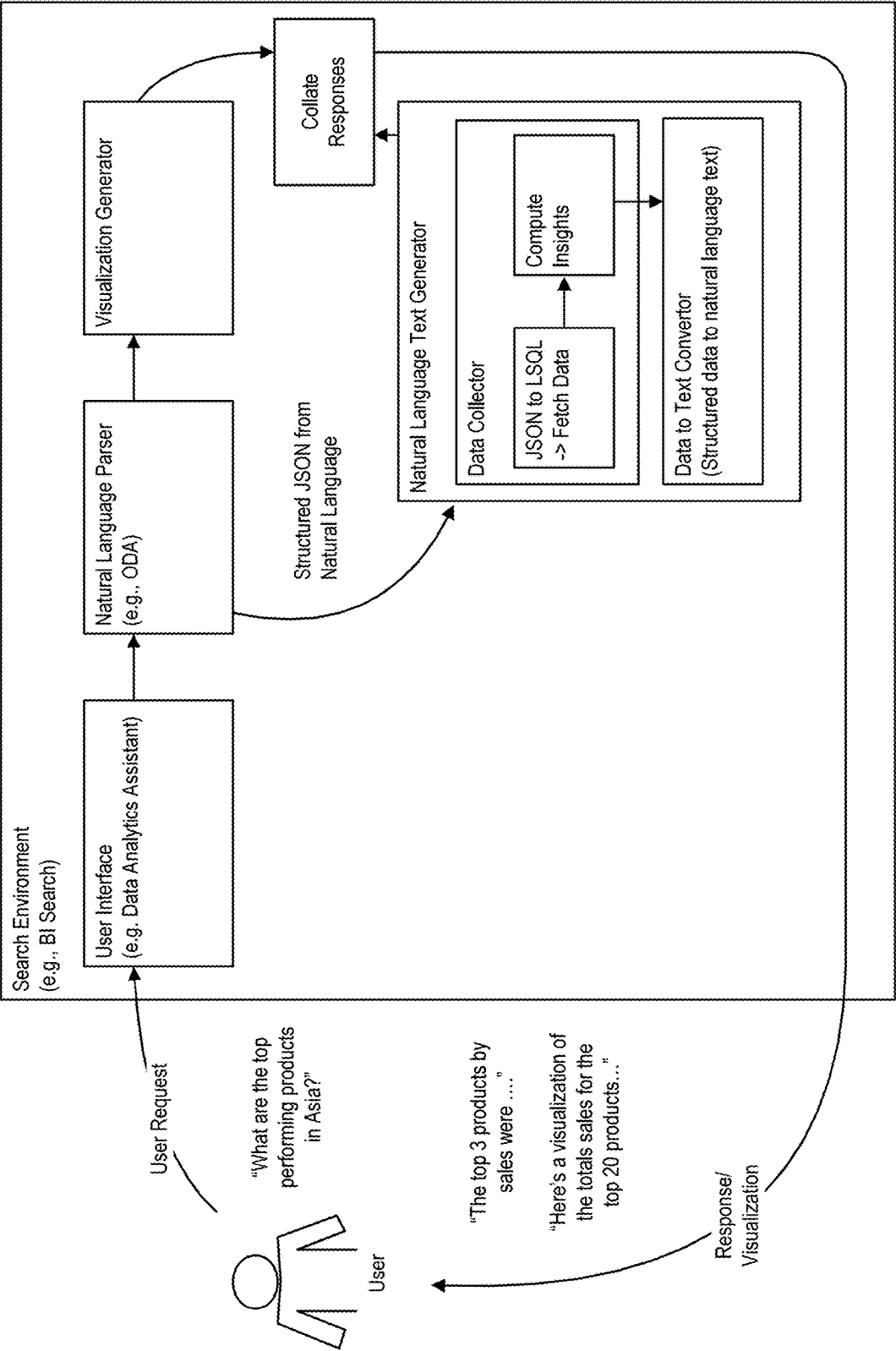


FIGURE 11

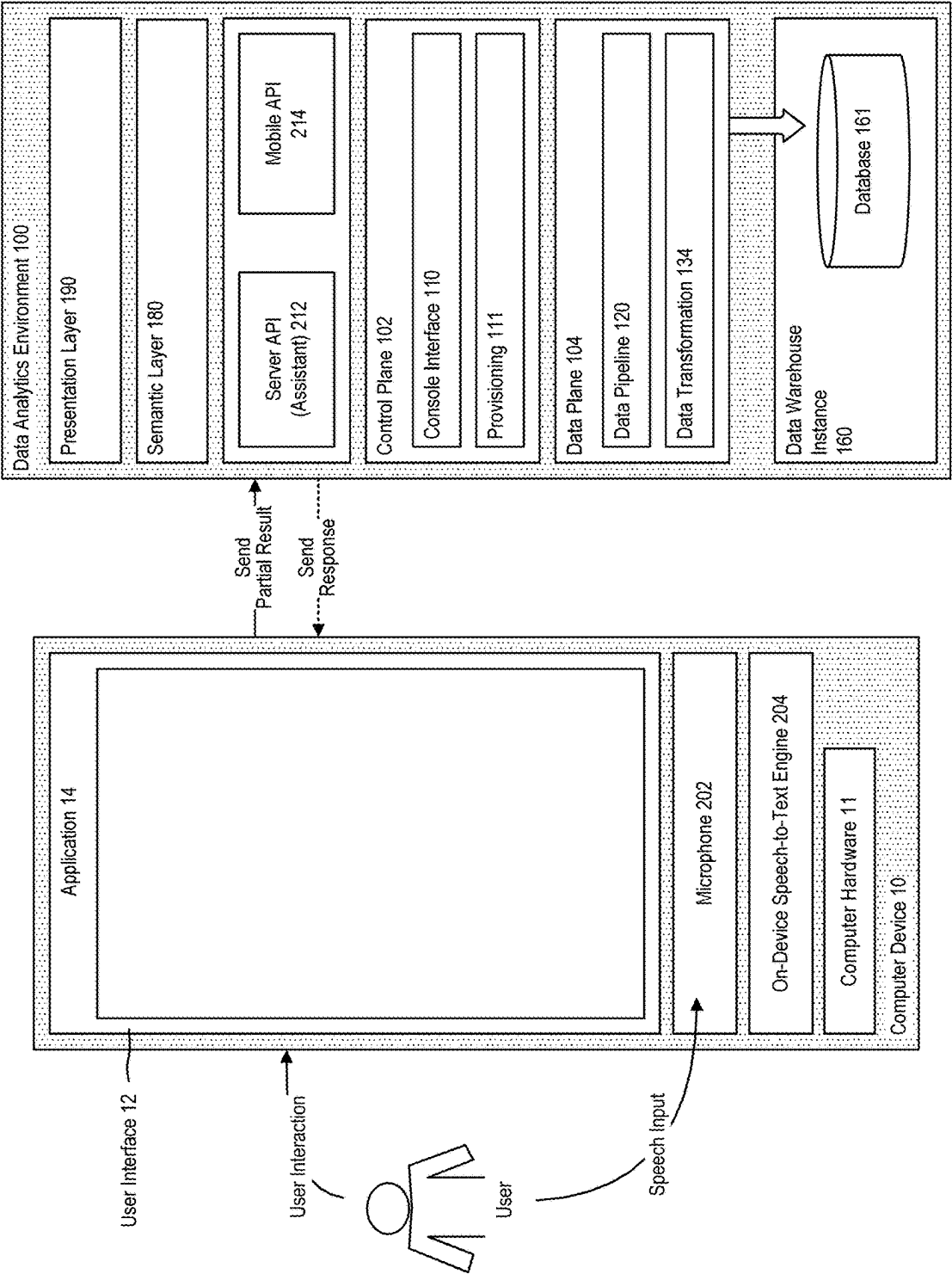


FIGURE 12A

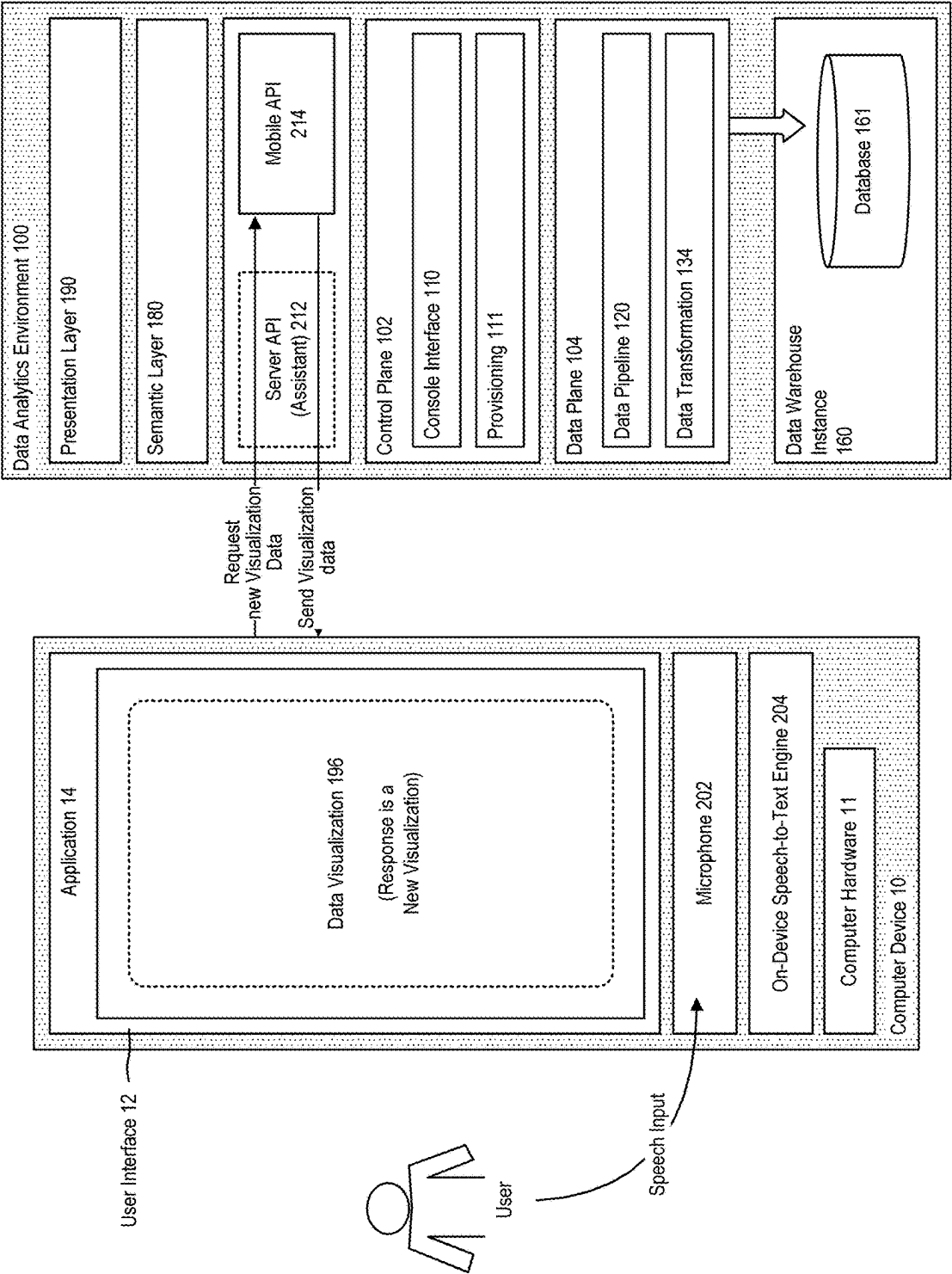


FIGURE 12B

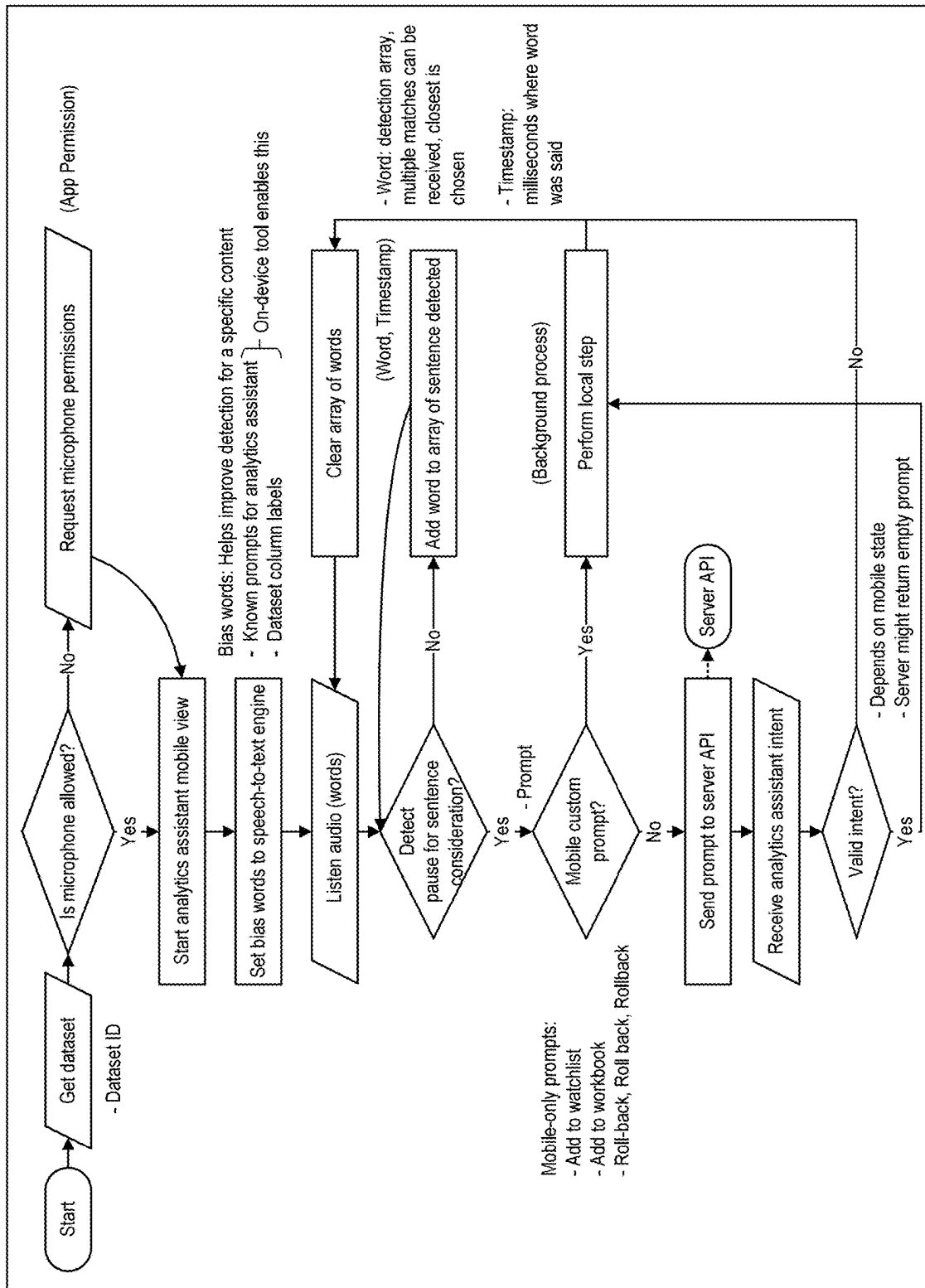


FIGURE 13

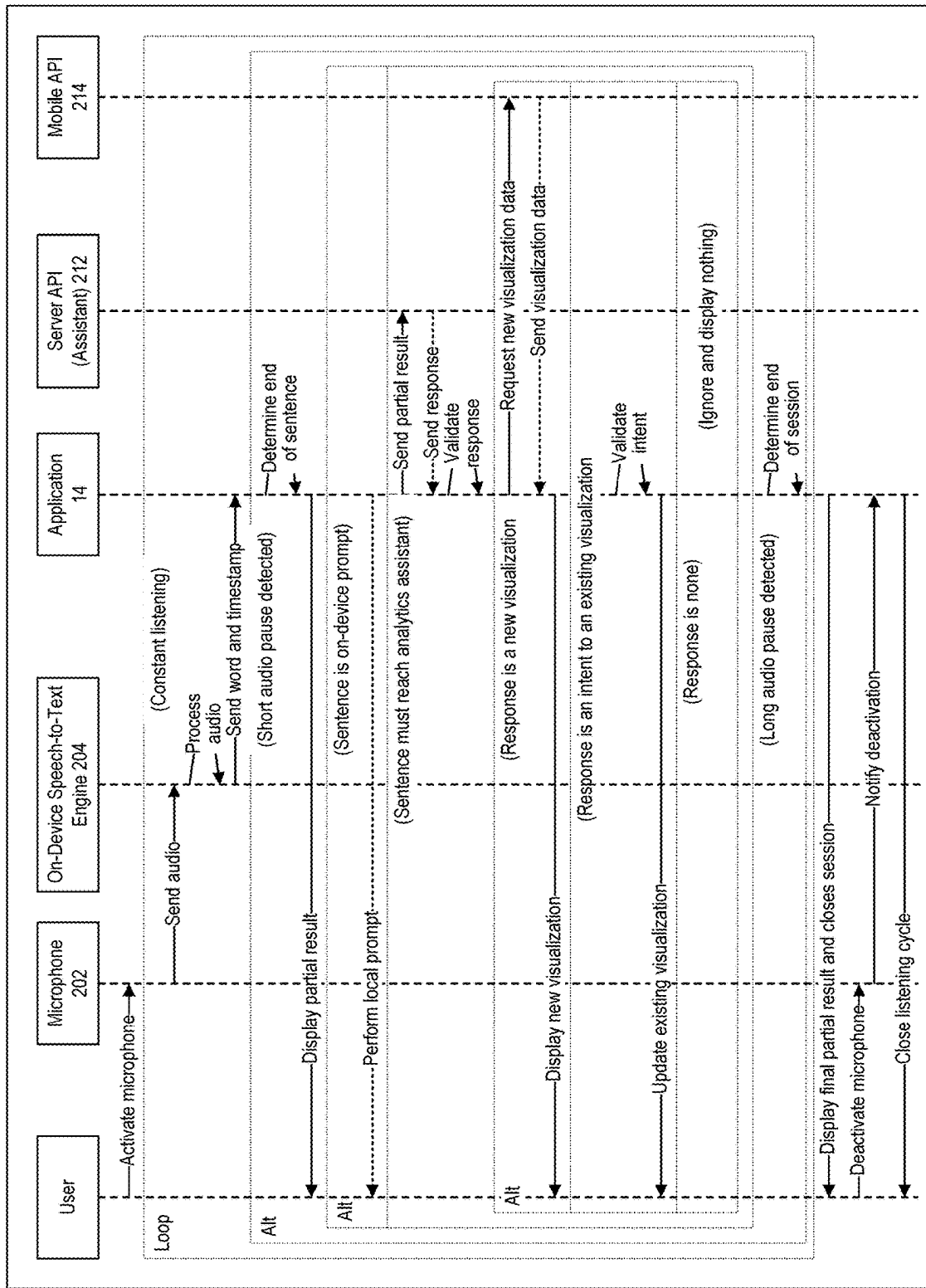


FIGURE 14

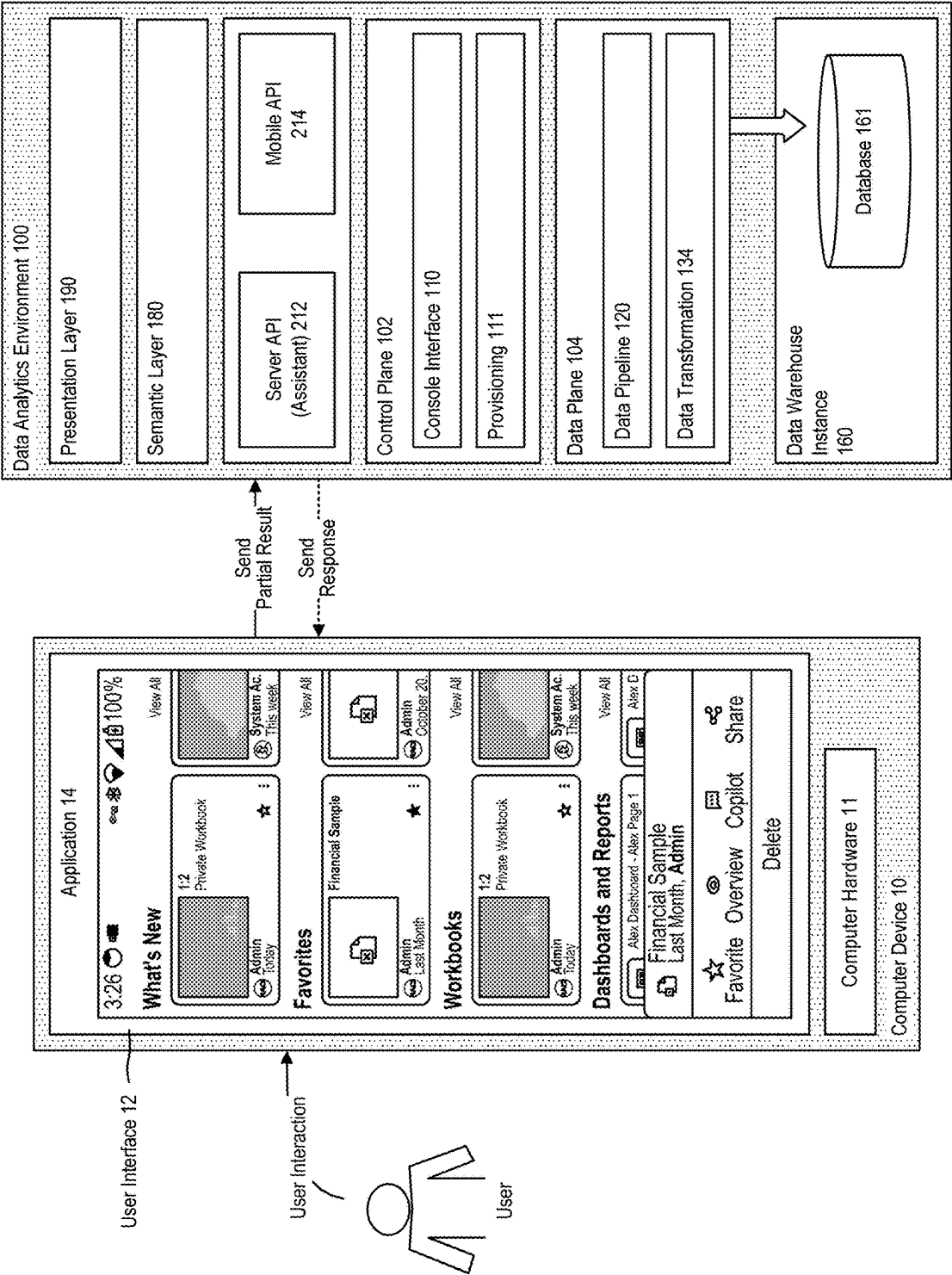


FIGURE 15A

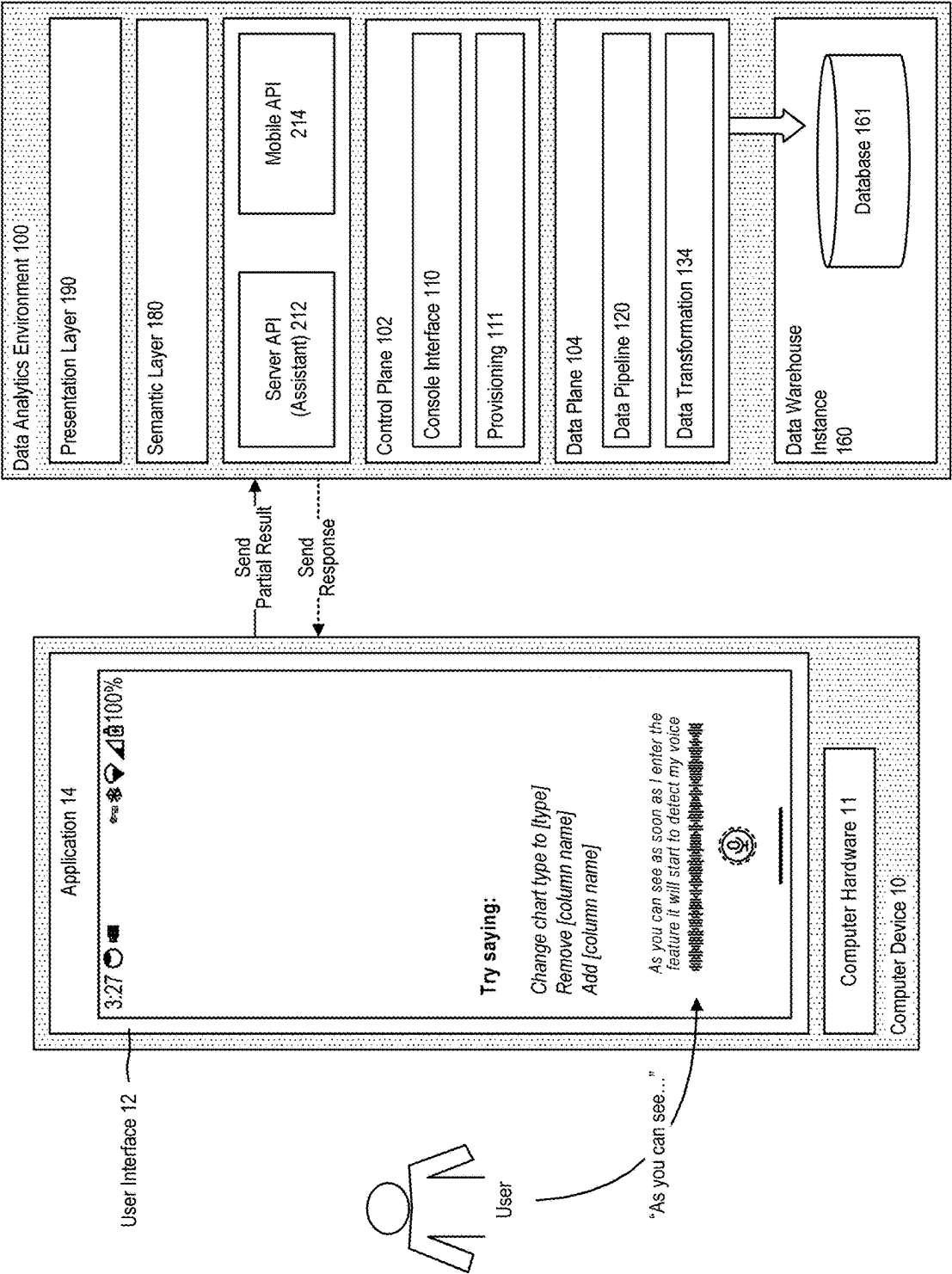


FIGURE 15B

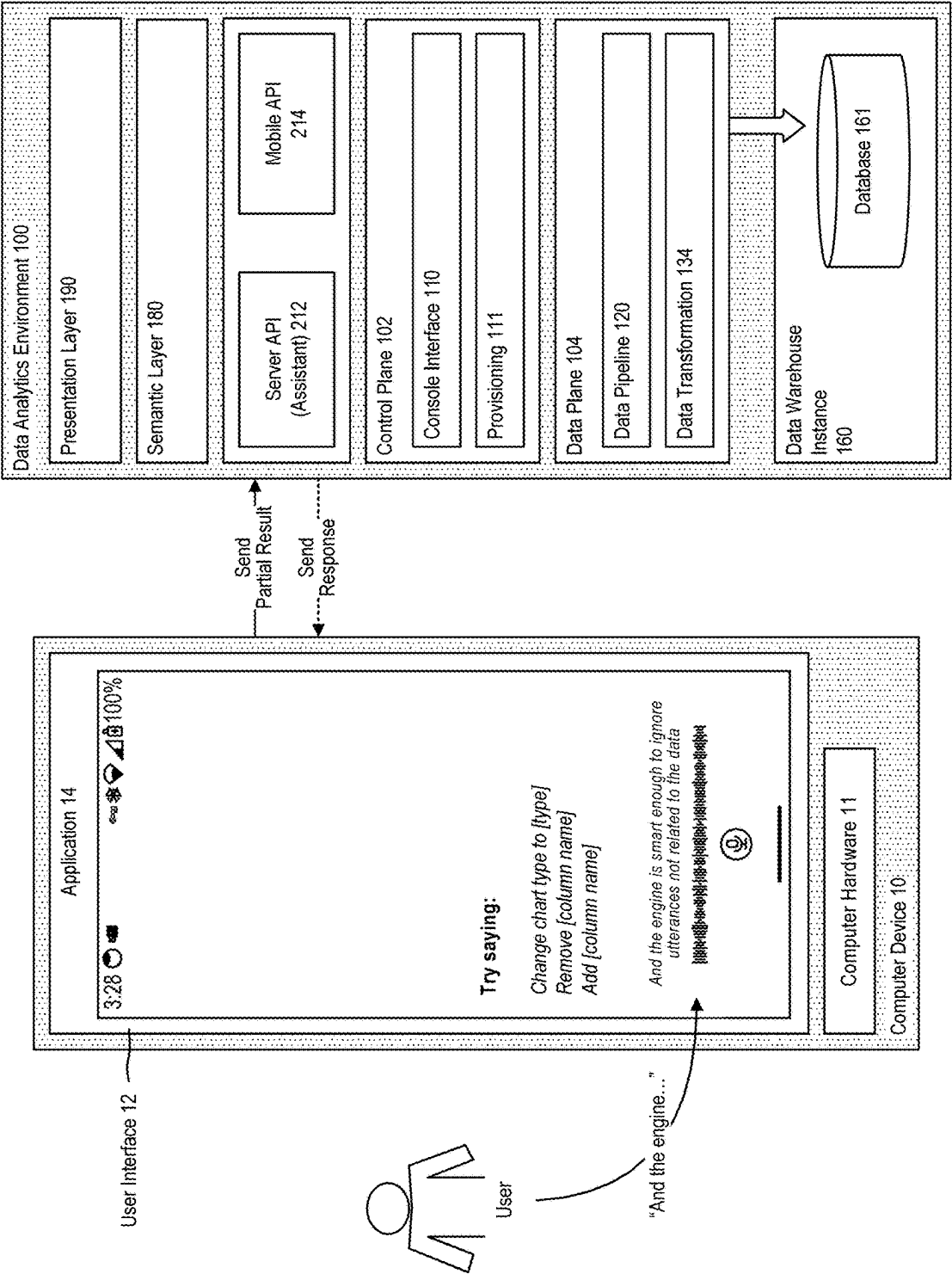


FIGURE 15C

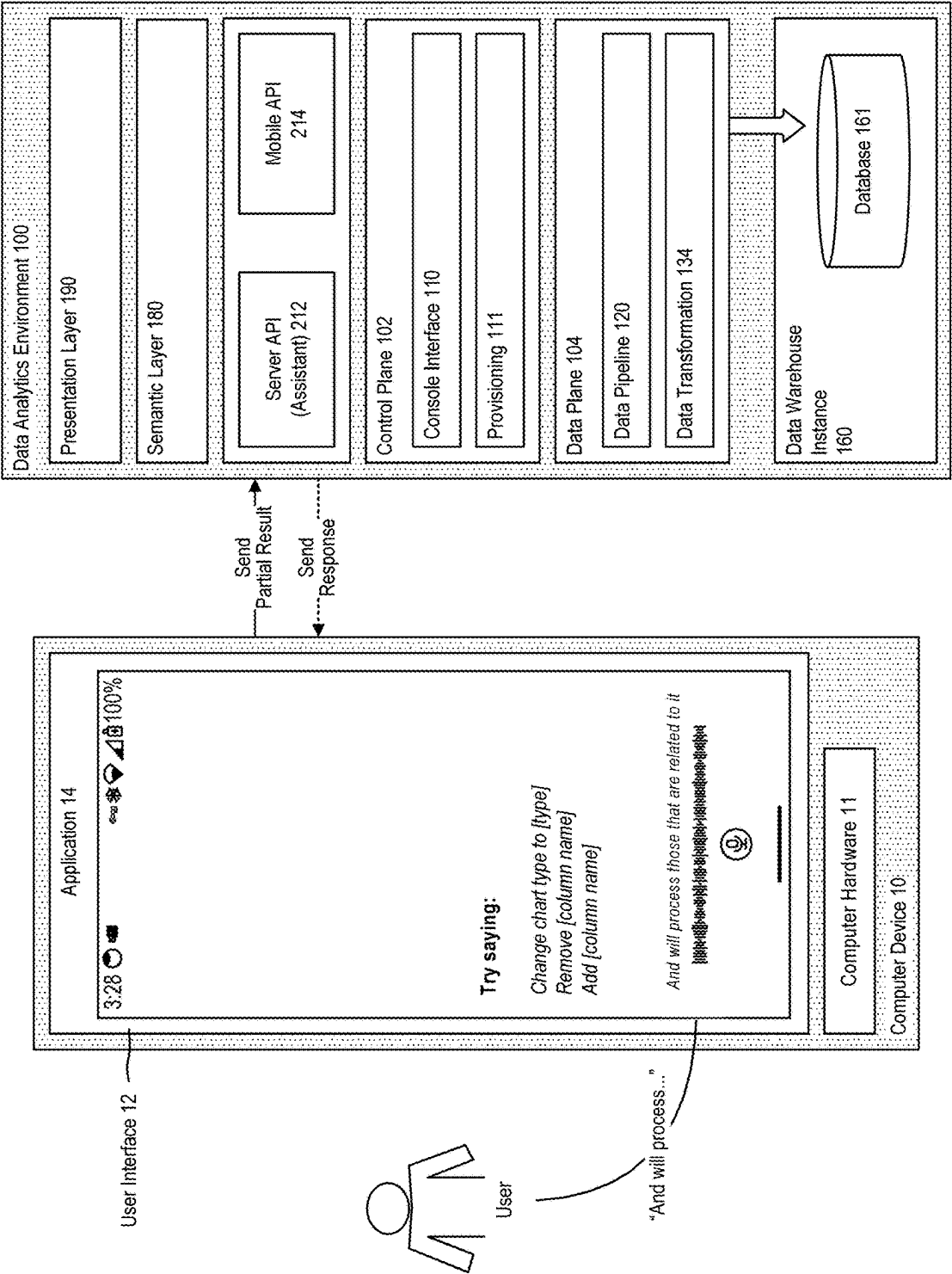


FIGURE 15D

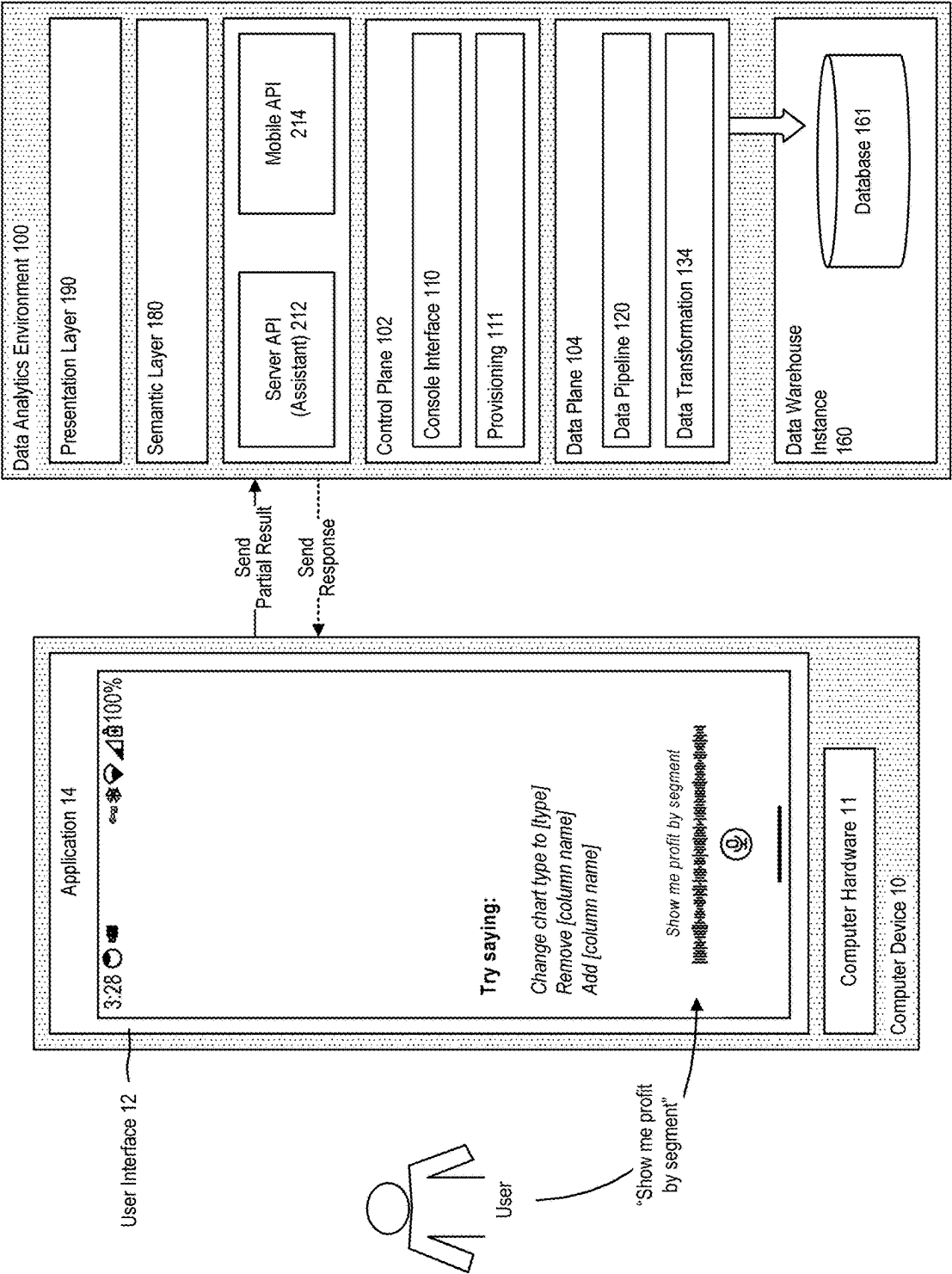


FIGURE 15E

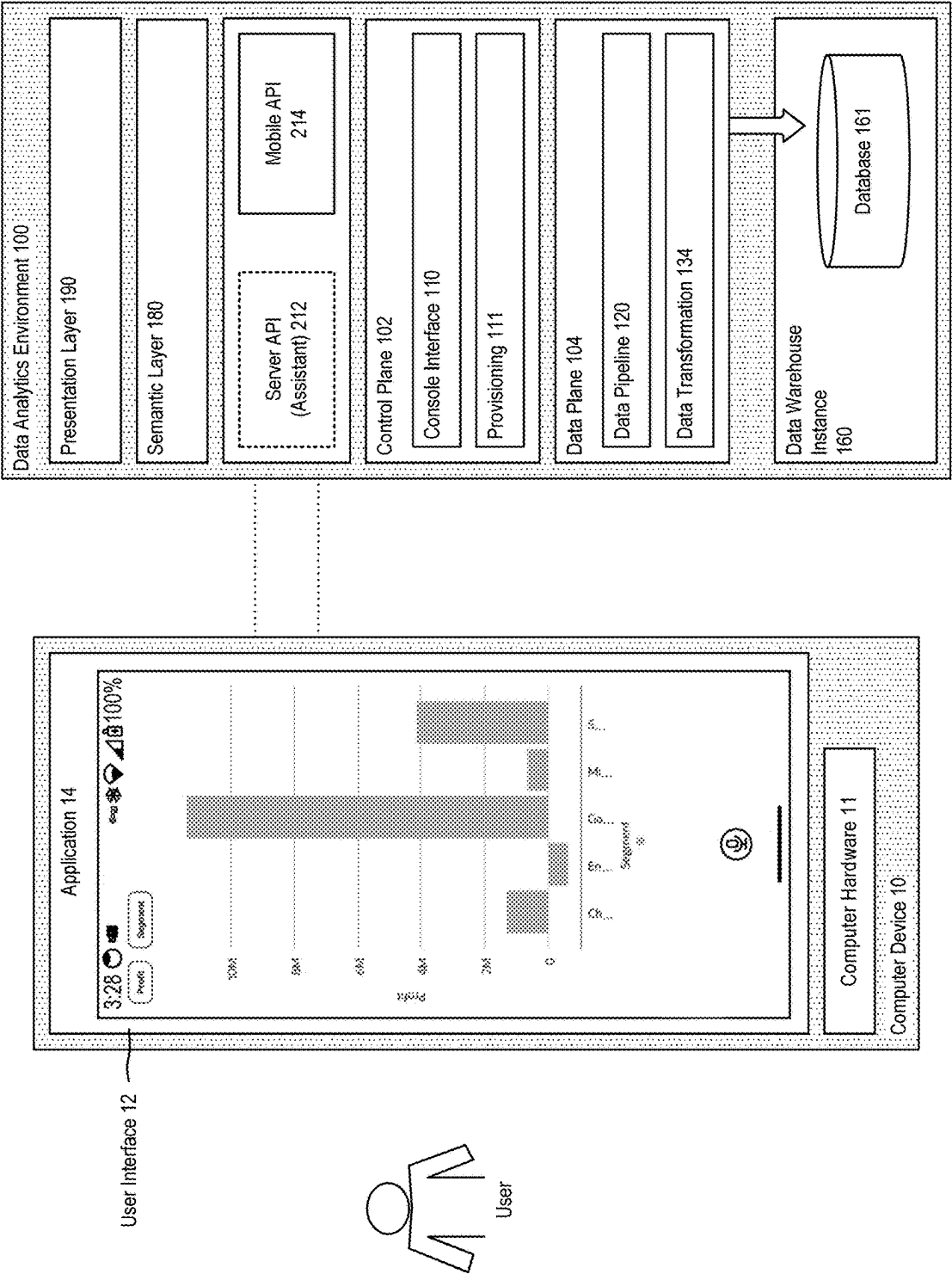


FIGURE 15F

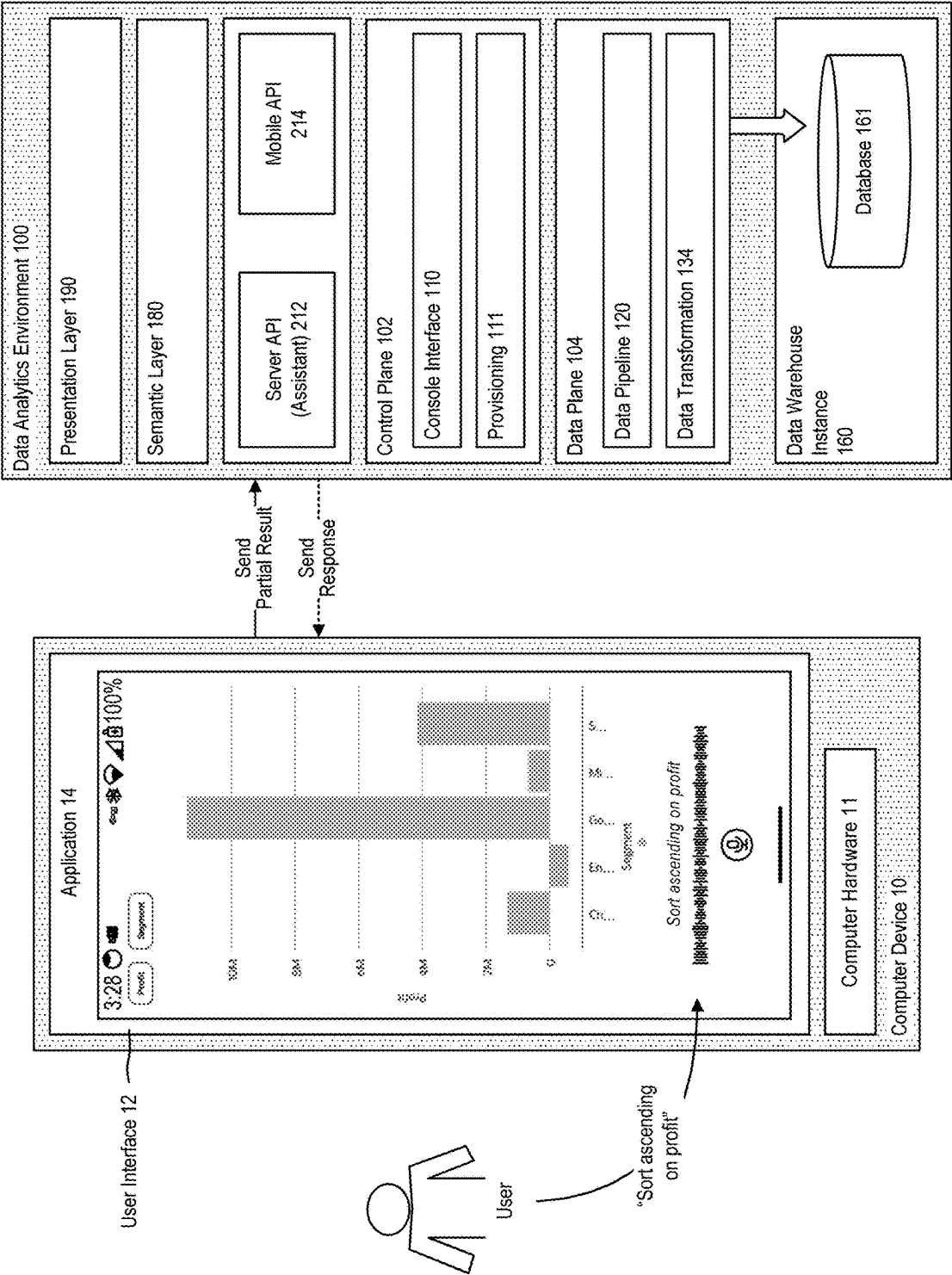


FIGURE 15G

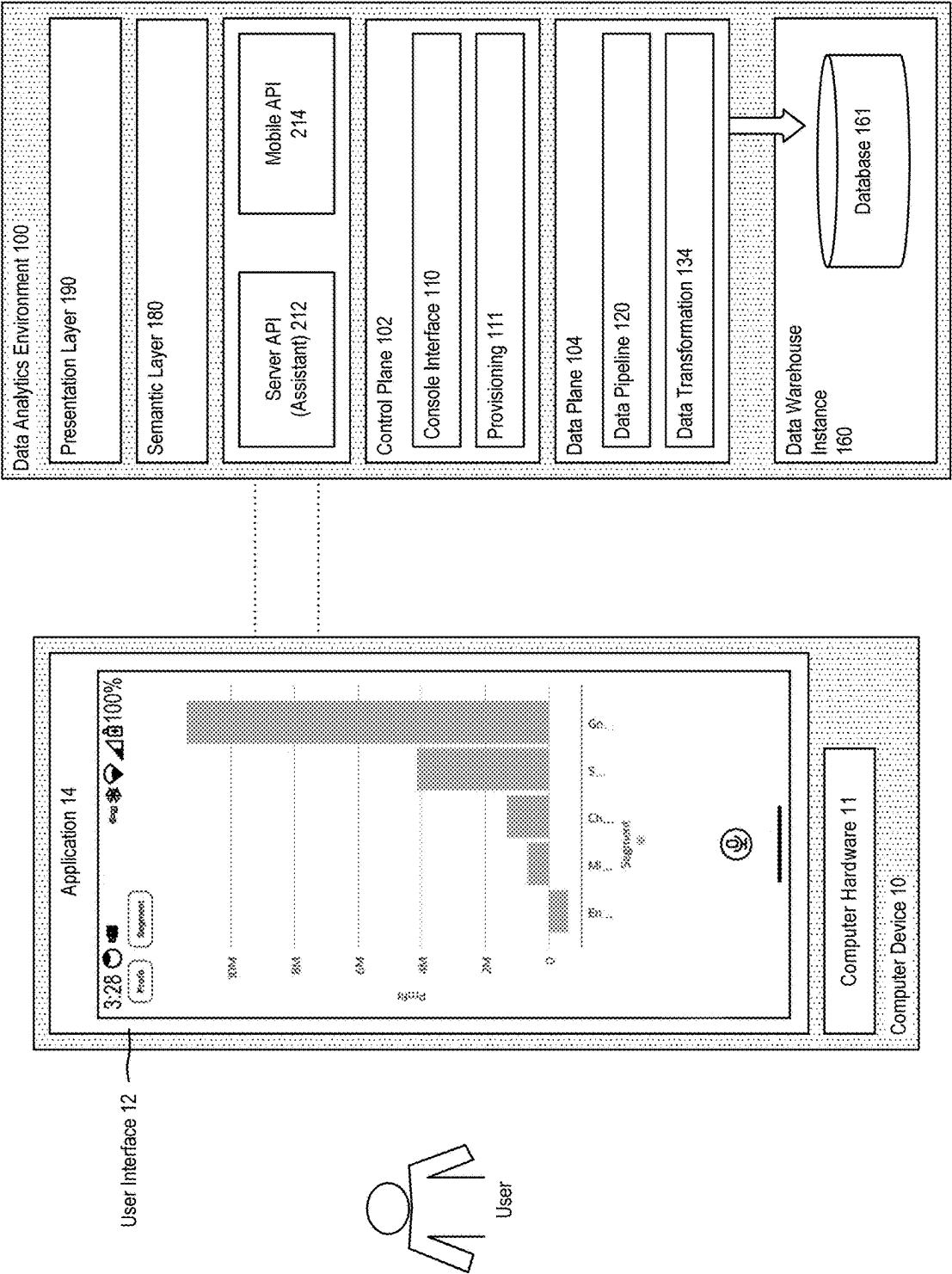


FIGURE 15H

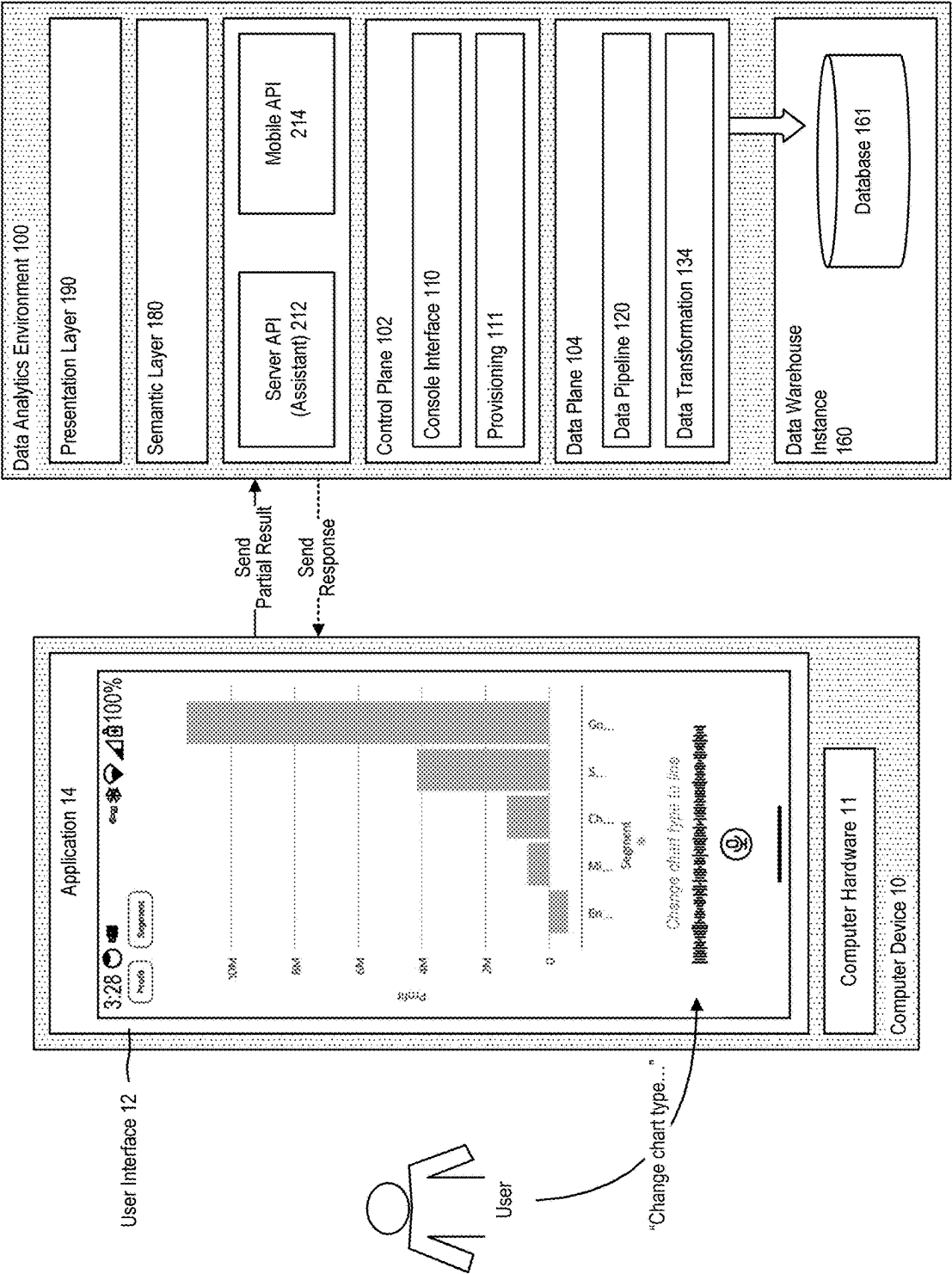


FIGURE 15I

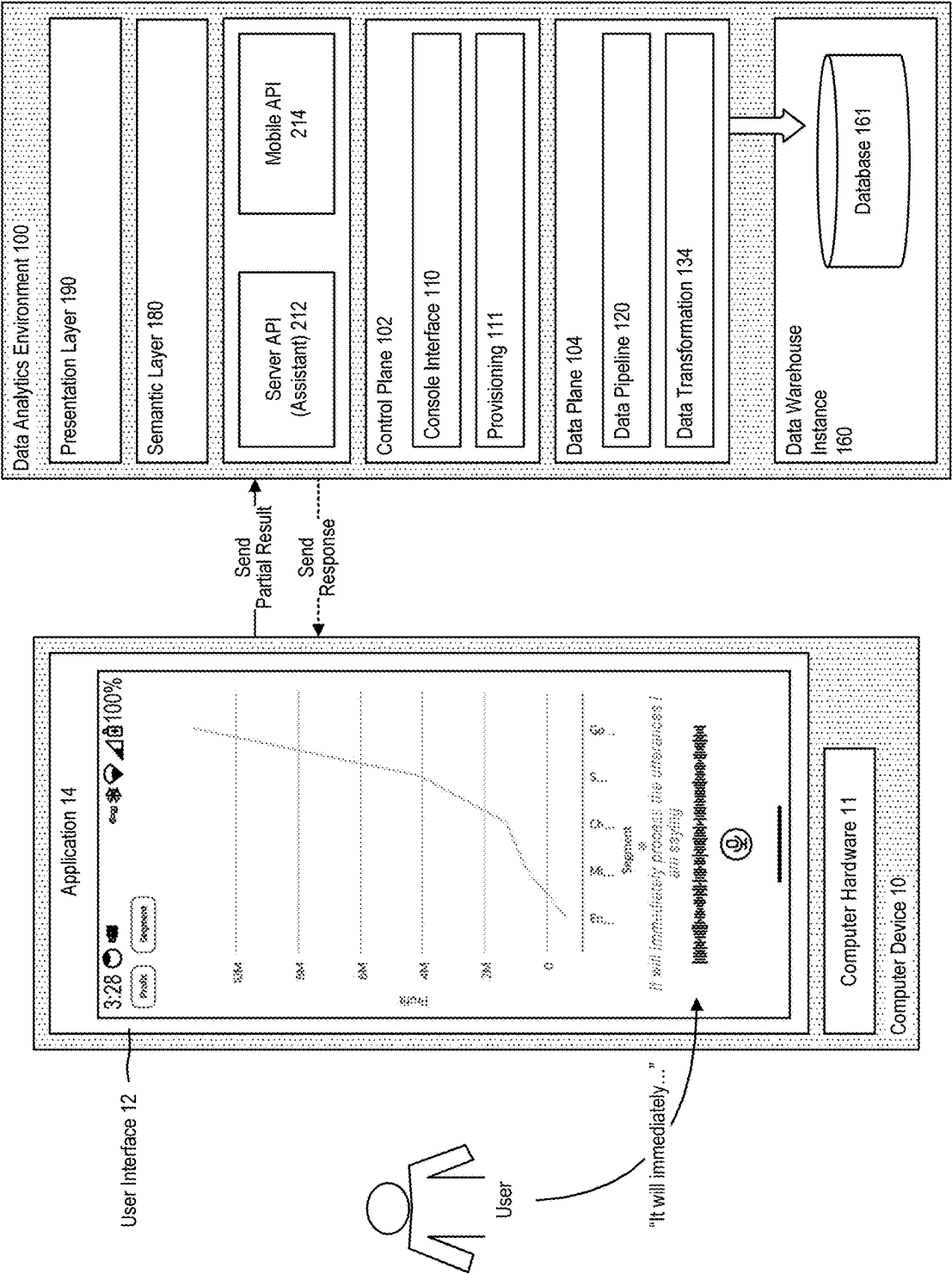


FIGURE 15J

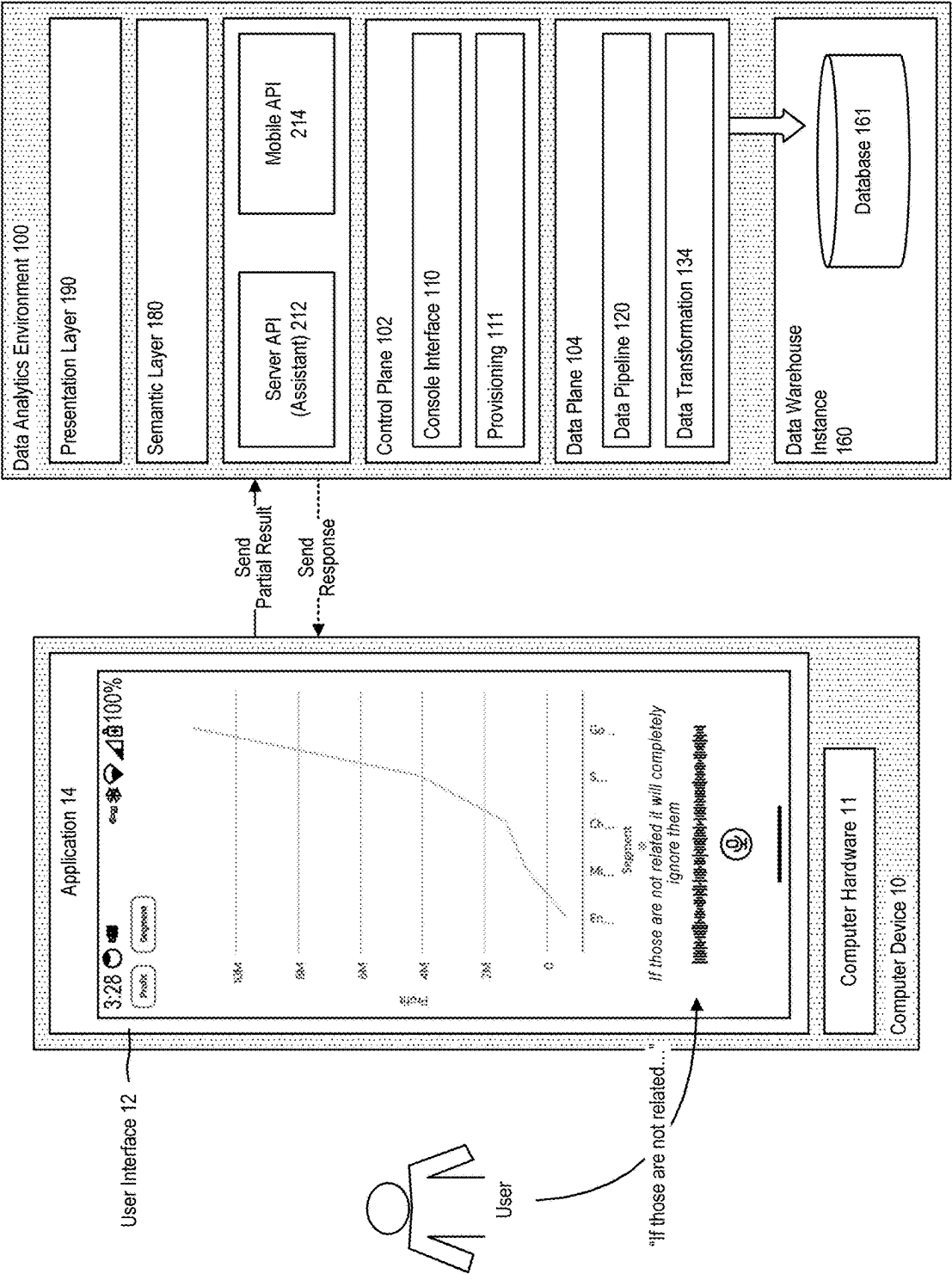


FIGURE 15K

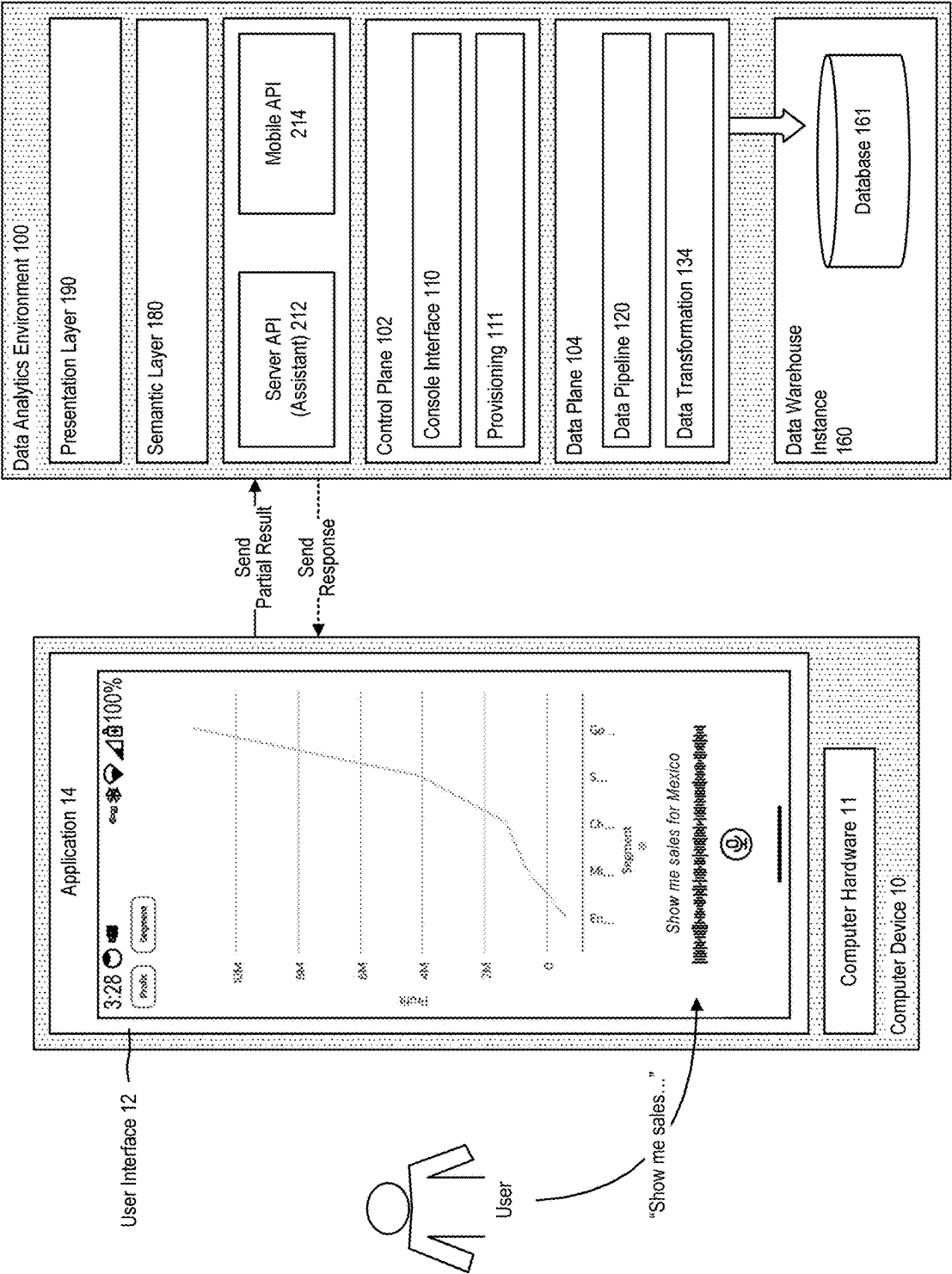


FIGURE 15L

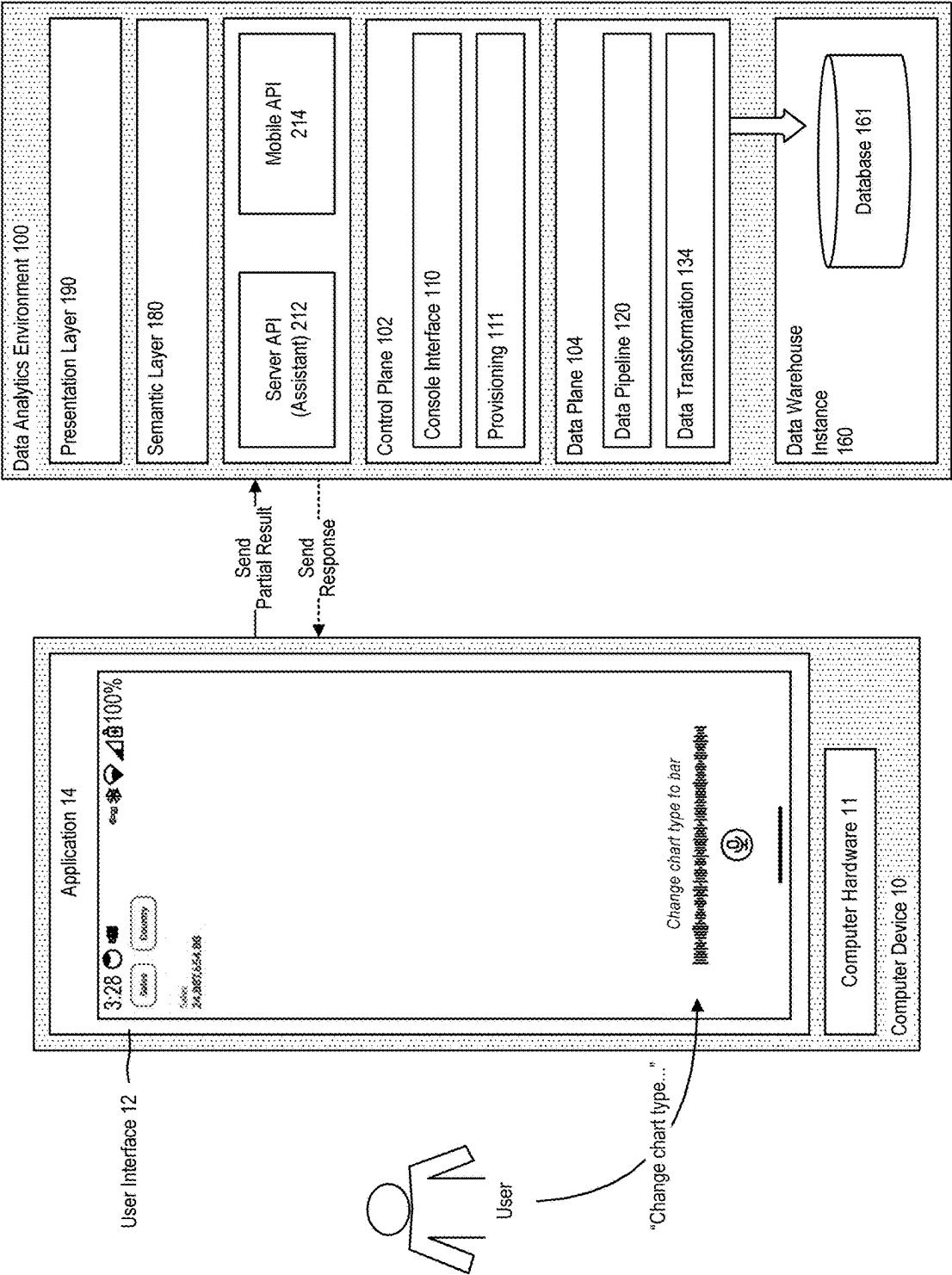


FIGURE 15M

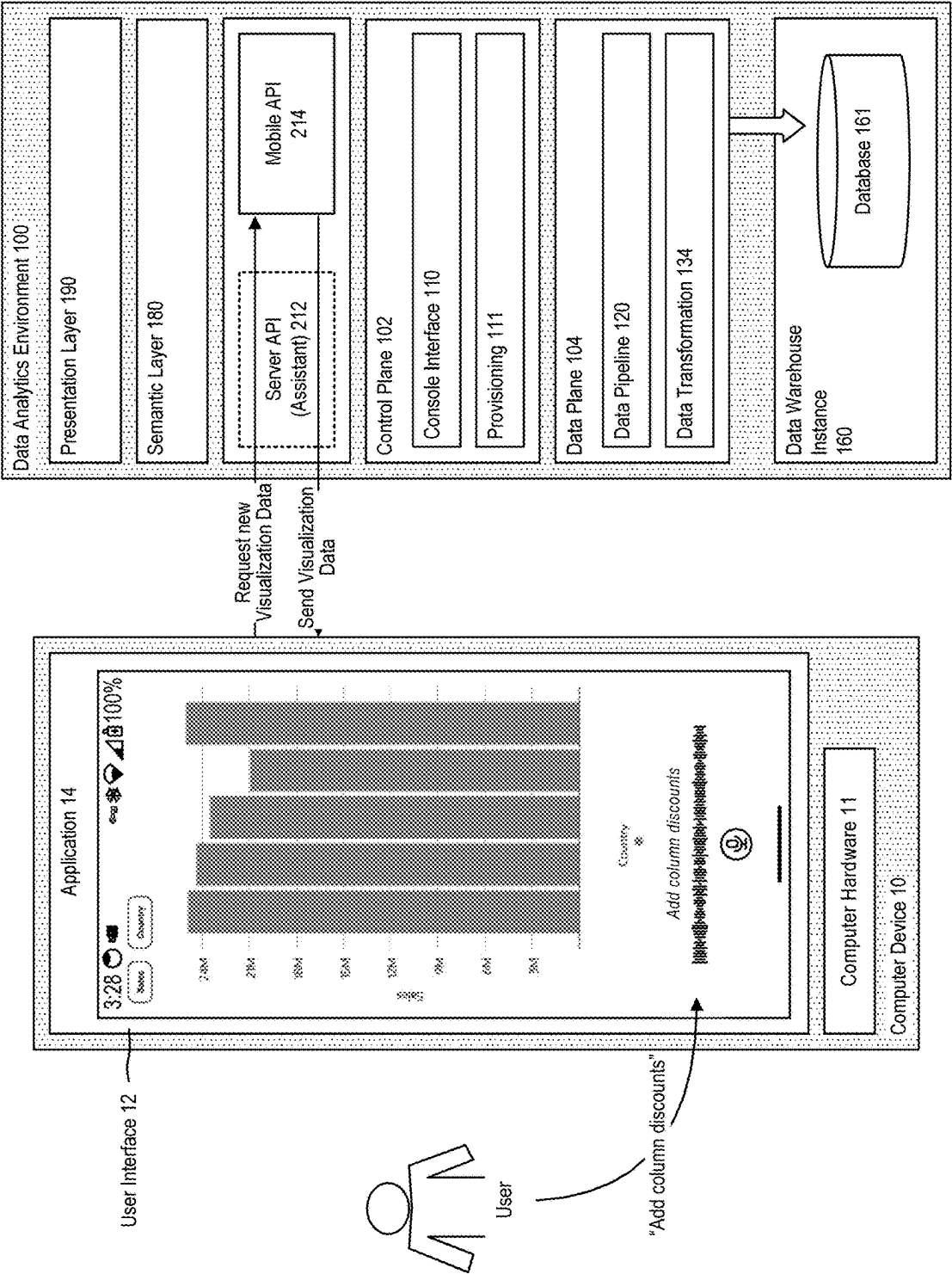


FIGURE 15N

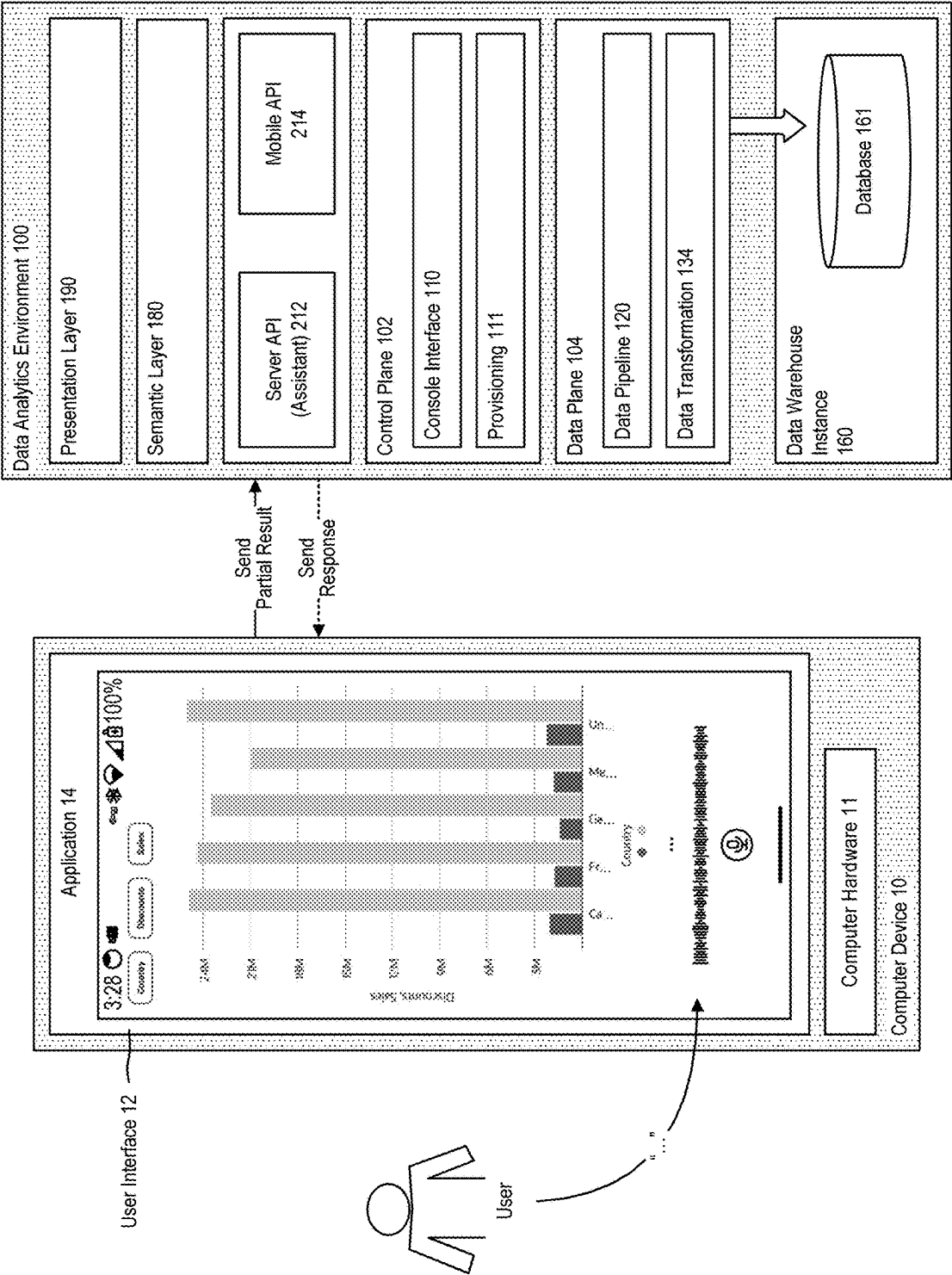


FIGURE 150

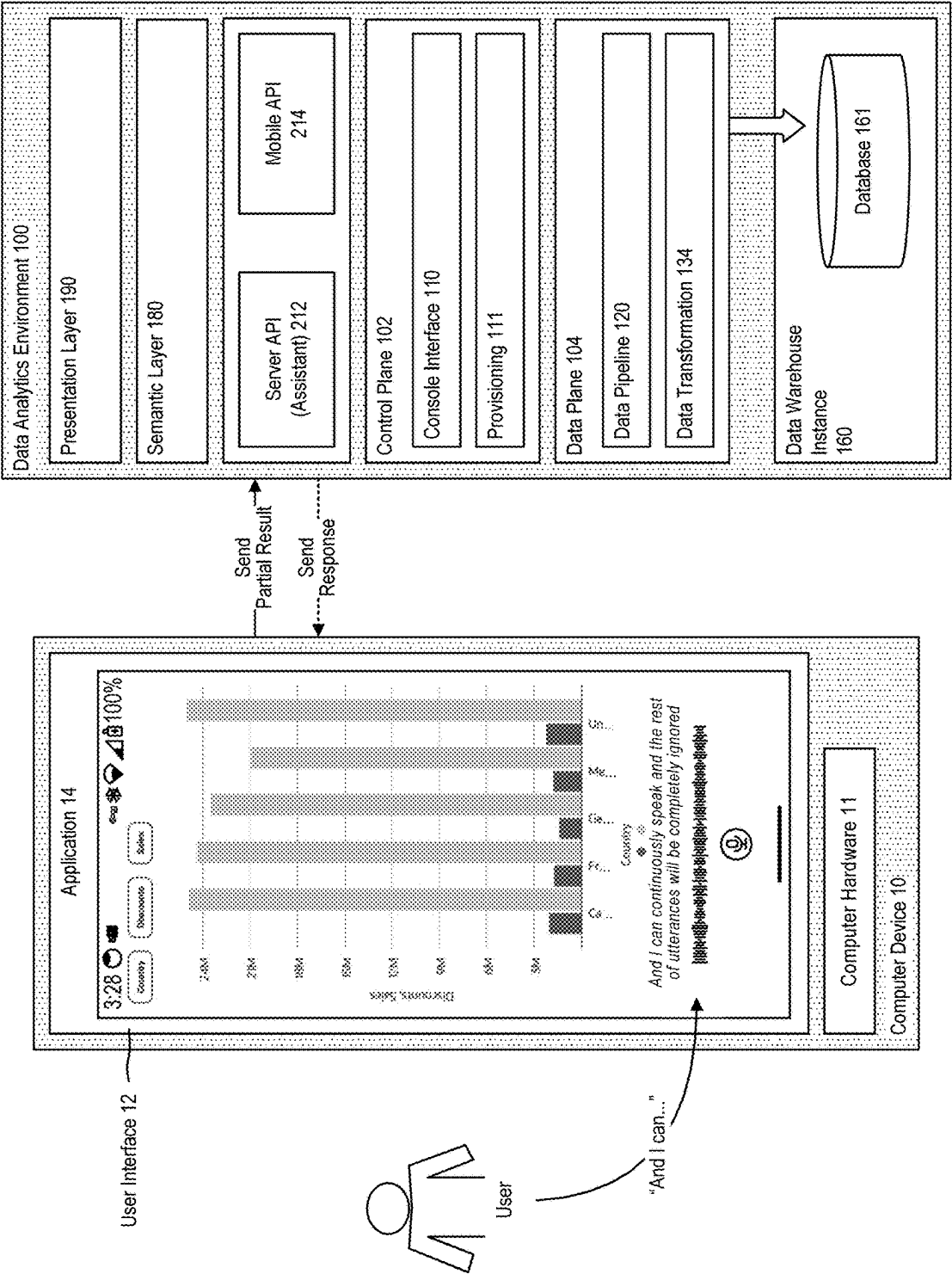


FIGURE 15P

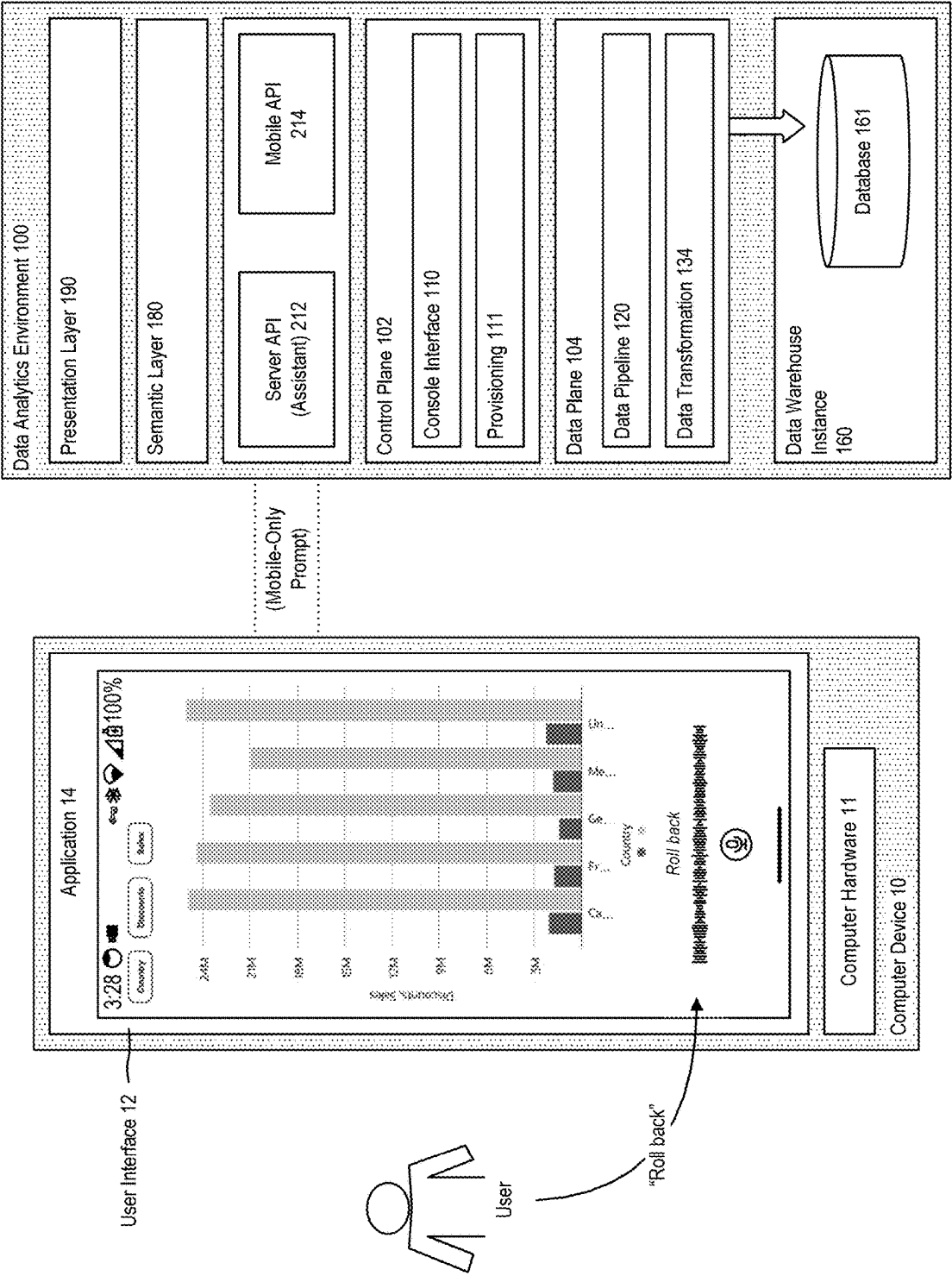


FIGURE 15Q

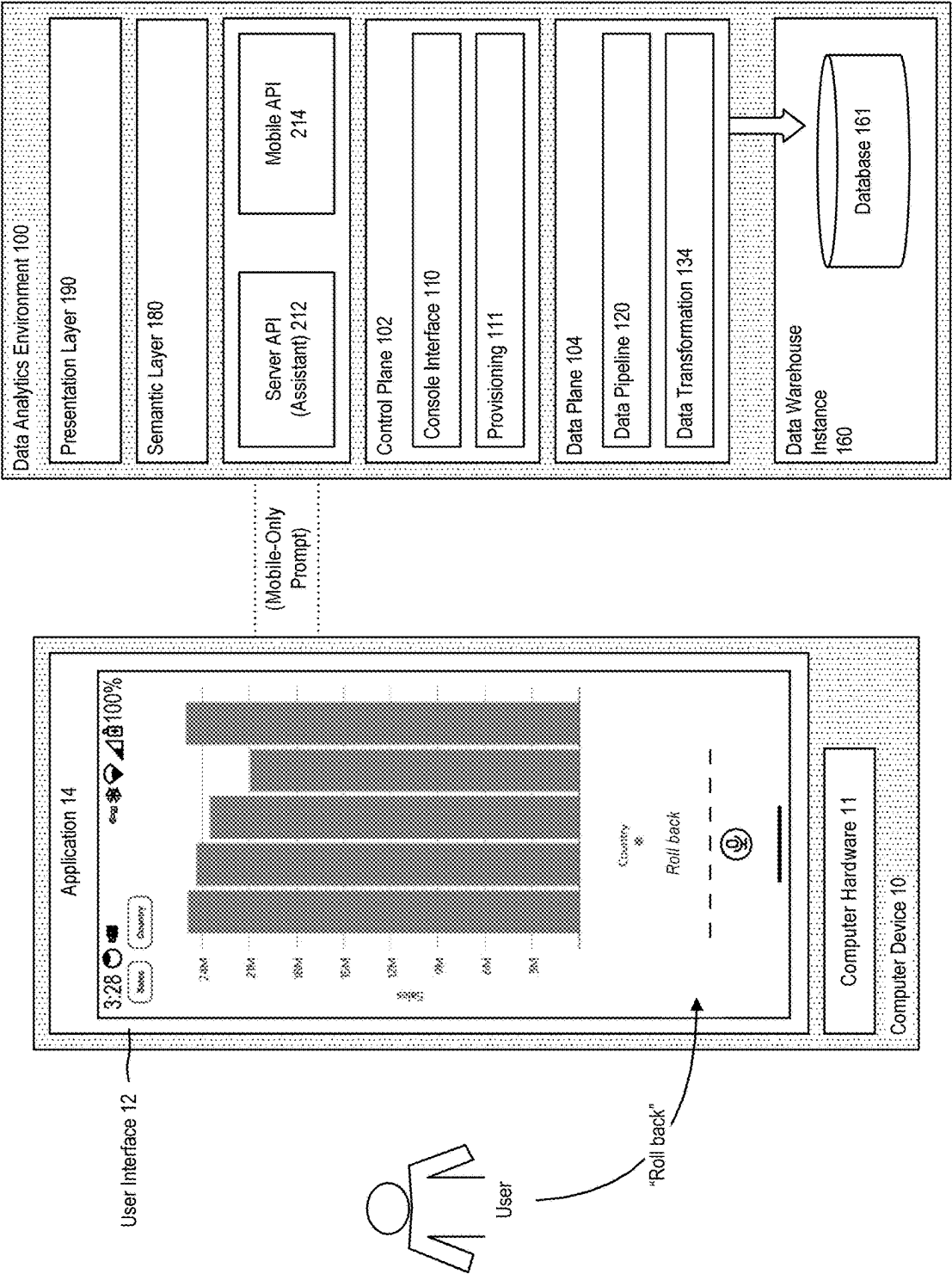


FIGURE 15R

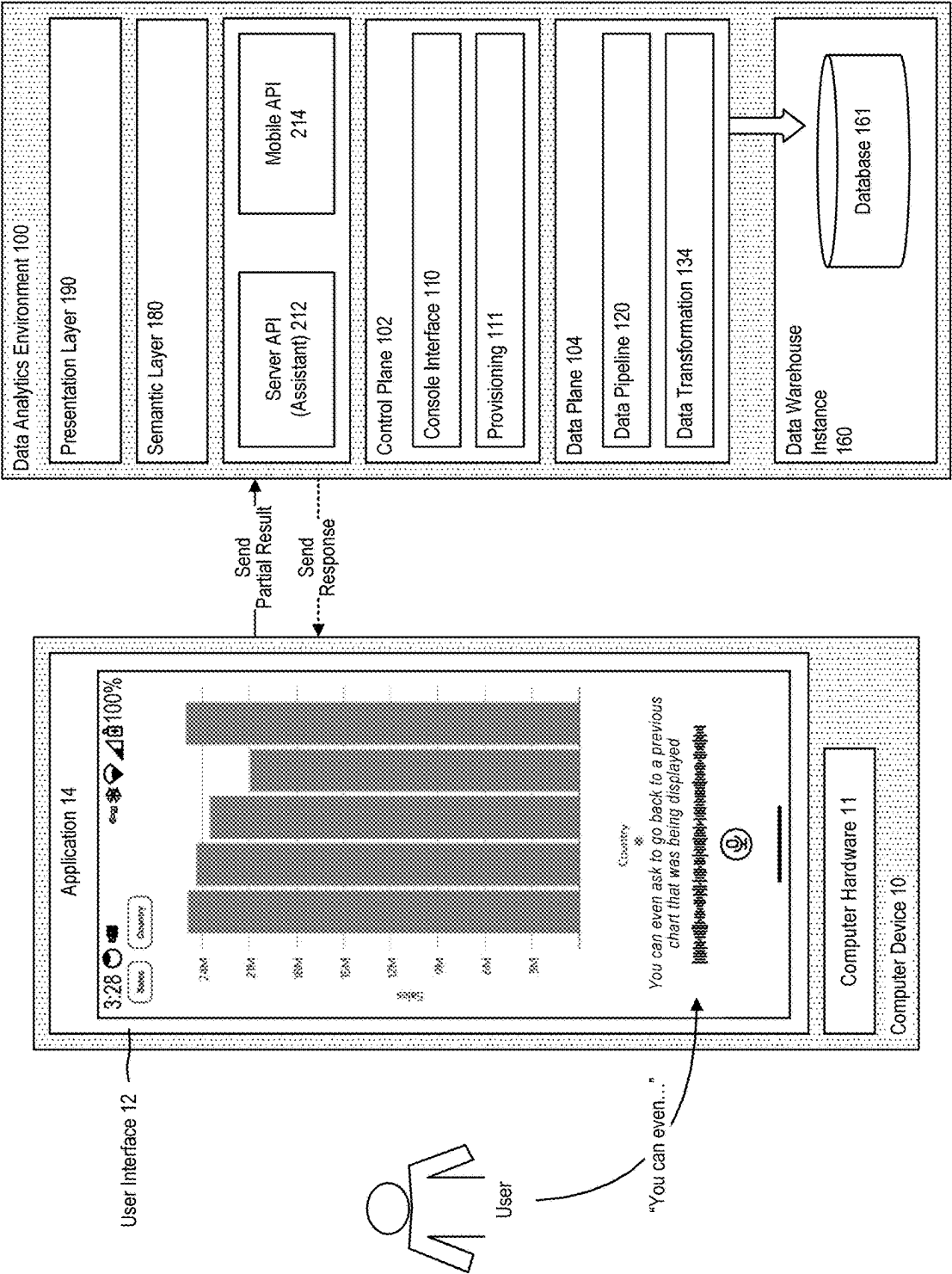


FIGURE 15S

**SYSTEM AND METHOD FOR PROVIDING
REAL-TIME SPEECH RECOGNITION AND
NATIVE VISUALIZATION FOR DATA
ANALYTICS**

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**CLAIM OF PRIORITY AND
CROSS-REFERENCE TO RELATED
APPLICATIONS**

[0002] This application claims the benefit of priority to U.S. Provisional patent application titled “SYSTEM AND METHOD FOR PROVIDING REAL-TIME SPEECH RECOGNITION AND NATIVE VISUALIZATION FOR DATA ANALYTICS”, Application No. 63/563,232, filed Mar. 8, 2024; and is related to U.S. patent application titled “TECHNIQUES FOR SEMANTIC SEARCHING”, application Ser. No. 16/662,695, filed Oct. 24, 2019, and published as U.S. Patent Application Publication No. 2020/0117658; each of which above applications and the contents thereof are herein incorporated by reference.

TECHNICAL FIELD

[0003] Embodiments described herein are generally related to computer data analytics, and computer-based methods of providing business intelligence or other data, and are particularly related to a system and method for providing a real-time speech recognition and native visualization tool, for use with data analytics environments.

BACKGROUND

[0004] Data analytics enables computer-based examination of large amounts of data, for example to derive conclusions or other information from the data. For example, business intelligence tools can be used to provide users with business intelligence describing their enterprise data, in a format that enables the users to make strategic business decisions.

SUMMARY

[0005] Embodiments described herein are generally related to computer data analytics, and computer-based methods of providing business intelligence or other data, and are particularly related to a system and method for providing a real-time speech recognition and native visualization tool, for use with data analytics environments. The described approach can be used to provide real-time data visualizations or other analytics based on a live conversation with a user using an on-device speech-to-text recognition engine.

[0006] In accordance with an embodiment, the method includes: listening for audio received as input from a user; building an array of words associated with a sentence; determining whether the sentence is capable of being processed without additional server data and if so performing

the step locally; and alternatively sending an input sentence to a server API, in order to determine an intent associated therewith for use by the server in returning relevant results.

BRIEF DESCRIPTION OF THE DRAWINGS

[0007] FIG. 1 illustrates an example data analytics environment, in accordance with an embodiment.

[0008] FIG. 2 further illustrates an example data analytics environment, in accordance with an embodiment.

[0009] FIG. 3 further illustrates an example data analytics environment, in accordance with an embodiment.

[0010] FIG. 4 further illustrates an example data analytics environment, in accordance with an embodiment.

[0011] FIG. 5 further illustrates an example data analytics environment, in accordance with an embodiment.

[0012] FIG. 6 illustrates a use of the system to transform, analyze, or visualize data, in accordance with an embodiment.

[0013] FIG. 7 illustrates the preparation of a data visualization for use with a data analytics environment, in accordance with an embodiment.

[0014] FIG. 8 further illustrates the preparation of a data visualization for use with a data analytics environment, in accordance with an embodiment.

[0015] FIG. 9 further illustrates the preparation of a data visualization for use with a data analytics environment, in accordance with an embodiment.

[0016] FIG. 10 illustrates a system for providing digital assistant integration with a data analytics assistant, in accordance with an embodiment.

[0017] FIG. 11 illustrates the use of a natural language generator service to support digital assistant integration, in accordance with an embodiment.

[0018] FIG. 12A illustrates the use of a tool for providing real-time speech recognition and native visualization for data analytics, in accordance with an embodiment.

[0019] FIG. 12B further illustrates the use of a tool for providing real-time speech recognition and native visualization for data analytics, in accordance with an embodiment.

[0020] FIG. 13 illustrates a process for providing real-time speech recognition and native visualization for data analytics, in accordance with an embodiment.

[0021] FIG. 14 further illustrates a process for providing real-time speech recognition and native visualization for data analytics, in accordance with an embodiment.

[0022] FIG. 15A illustrates an example use of a tool for providing real-time speech recognition and native visualization for data analytics, in accordance with an embodiment.

[0023] FIG. 15B further illustrates an example use of a tool for providing real-time speech recognition and native visualization for data analytics, in accordance with an embodiment.

[0024] FIG. 15C further illustrates an example use of a tool for providing real-time speech recognition and native visualization for data analytics, in accordance with an embodiment.

[0025] FIG. 15D further illustrates an example use of a tool for providing real-time speech recognition and native visualization for data analytics, in accordance with an embodiment.

[0026] FIG. 15E further illustrates an example use of a tool for providing real-time speech recognition and native visualization for data analytics, in accordance with an embodiment.

[0027] FIG. 15F further illustrates an example use of a tool for providing real-time speech recognition and native visualization for data analytics, in accordance with an embodiment.

[0028] FIG. 15G further illustrates an example use of a tool for providing real-time speech recognition and native visualization for data analytics, in accordance with an embodiment.

[0029] FIG. 15H further illustrates an example use of a tool for providing real-time speech recognition and native visualization for data analytics, in accordance with an embodiment.

[0030] FIG. 15I further illustrates an example use of a tool for providing real-time speech recognition and native visualization for data analytics, in accordance with an embodiment.

[0031] FIG. 15J further illustrates an example use of a tool for providing real-time speech recognition and native visualization for data analytics, in accordance with an embodiment.

[0032] FIG. 15K further illustrates an example use of a tool for providing real-time speech recognition and native visualization for data analytics, in accordance with an embodiment.

[0033] FIG. 15L further illustrates an example use of a tool for providing real-time speech recognition and native visualization for data analytics, in accordance with an embodiment.

[0034] FIG. 15M further illustrates an example use of a tool for providing real-time speech recognition and native visualization for data analytics, in accordance with an embodiment.

[0035] FIG. 15N further illustrates an example use of a tool for providing real-time speech recognition and native visualization for data analytics, in accordance with an embodiment.

[0036] FIG. 15O further illustrates an example use of a tool for providing real-time speech recognition and native visualization for data analytics, in accordance with an embodiment.

[0037] FIG. 15P further illustrates an example use of a tool for providing real-time speech recognition and native visualization for data analytics, in accordance with an embodiment.

[0038] FIG. 15Q further illustrates an example use of a tool for providing real-time speech recognition and native visualization for data analytics, in accordance with an embodiment.

[0039] FIG. 15R further illustrates an example use of a tool for providing real-time speech recognition and native visualization for data analytics, in accordance with an embodiment.

[0040] FIG. 15S further illustrates an example use of a tool for providing real-time speech recognition and native visualization for data analytics, in accordance with an embodiment.

DETAILED DESCRIPTION

[0041] Generally described, within an organization, data analytics enables computer-based examination of large amounts of data, for example to derive conclusions or other information from the data. For example, business intelligence (BI) tools can be used to provide users with business

intelligence describing their enterprise data, in a format that enables the users to make strategic business decisions.

[0042] Increasingly, data analytics can be provided within the context of enterprise software application environments, such as, for example, an Oracle Fusion Applications environment; or within the context of software-as-a-service (SaaS) or cloud environments, such as, for example, an Oracle Analytics Cloud or Oracle Cloud Infrastructure environment; or other types of analytics application or cloud environments.

[0043] Examples of data analytics environments and business intelligence tools/servers include Oracle Business Intelligence Server (OBIS), Oracle Analytics Cloud (OAC), and Fusion Analytics Warehouse (FAW), which support features such as data mining or analytics, and analytic applications.

[0044] FIG. 1 illustrates an example data analytics environment, in accordance with an embodiment.

[0045] The example embodiment illustrated in FIG. 1 is provided for purposes of illustrating an example of a data analytics environment in association with which various embodiments described herein can be used. In accordance with other embodiments and examples, the approach described herein can be used with other types of data analytics, database, or data warehouse environments. The components and processes illustrated in FIG. 1, and as further described herein with regard to various other embodiments, can be provided as software or program code executable by, for example, a cloud computing system, or other suitably-programmed computer system.

[0046] As illustrated in FIG. 1, in accordance with an embodiment, a data analytics environment 100 can be provided by, or otherwise operate at, a computer system having a computer hardware (e.g., processor, memory) 101, and including one or more software components operating as a control plane 102, and a data plane 104, and providing access to a data warehouse, data warehouse instance 160 (database 161, or other type of data source).

[0047] In accordance with an embodiment, the control plane operates to provide control for cloud or other software products offered within the context of a SaaS or cloud environment, such as, for example, an Oracle Analytics Cloud environment, or other type of cloud environment. For example, in accordance with an embodiment, the control plane can include a console interface 110 that enables access by a customer (tenant) and/or a cloud environment having a provisioning component 111.

[0048] In accordance with an embodiment, the console interface can enable access by a customer (tenant) operating a graphical user interface (GUI) and/or a command-line interface (CLI) or other interface; and/or can include interfaces for use by providers of the SaaS or cloud environment and its customers (tenants). For example, in accordance with an embodiment, the console interface can provide interfaces that allow customers to provision services for use within their SaaS environment, and to configure those services that have been provisioned.

[0049] In accordance with an embodiment, a customer (tenant) can request via the console interface, a number of attributes associated with the data warehouse instance, including required attributes (e.g., login credentials), and optional attributes (e.g., size, or speed). The provisioning component can then provision the requested data warehouse instance, including a customer schema of the data warehouse; and populate the data warehouse instance with the

appropriate information supplied by the customer. The provisioning component can also be used to update or edit a data warehouse instance, and/or an ETL process that operates at the data plane, for example, by altering or updating a requested frequency of ETL process runs, for a particular customer (tenant).

[0050] In accordance with an embodiment, the data plane can include a data pipeline or process layer **120** and a data transformation layer **134**, that together process operational or transactional data from an organization's enterprise software application or data environment, such as, for example, business productivity software applications provisioned in a customer's (tenant's) SaaS environment. The data pipeline or process can include various functionality that extracts transactional data from business applications and databases that are provisioned in the SaaS environment, and then load a transformed data into the data warehouse.

[0051] In accordance with an embodiment, the data transformation layer can include a data model, such as, for example, a knowledge model (KM), or other type of data model, that the system uses to transform the transactional data received from business applications and corresponding transactional databases provisioned in the SaaS environment, into a model format understood by the data analytics environment.

[0052] In accordance with an embodiment, the data plane is responsible for performing extract, transform, and load (ETL) operations, including extracting transactional data from an organization's enterprise software application or data environment, such as, for example, business productivity software applications and corresponding transactional databases offered in a SaaS environment, transforming the extracted data into a model format, and loading the transformed data into a customer schema of the data warehouse.

[0053] For example, in accordance with an embodiment, each customer (tenant) of the environment can be associated with their own customer tenancy within the data warehouse, that is associated with their own customer schema; and can be additionally provided with read-only access to the data analytics schema, which can be updated by a data pipeline or process, for example, an ETL process, on a periodic or other basis.

[0054] In accordance with an embodiment, a data pipeline or process can be scheduled to execute at intervals (e.g., hourly/daily/weekly) to extract transactional data from an enterprise software application or data environment, such as, for example, business productivity software applications and corresponding transactional databases **106** that are provisioned in the SaaS environment.

[0055] In accordance with an embodiment, an extract process **108** can extract the transactional data, whereupon extraction the data pipeline or process can insert extracted data into a data staging area, which can act as a temporary staging area for the extracted data. The data quality component and data protection component can be used to ensure the integrity of the extracted data. For example, in accordance with an embodiment, the data quality component can perform validations on the extracted data while the data is temporarily held in the data staging area.

[0056] In accordance with an embodiment, when the extract process has completed its extraction, the data transformation layer can be used to begin the transform process, to transform the extracted data into a model format to be loaded into the customer schema of the data warehouse.

[0057] In accordance with an embodiment, the data pipeline or process can operate in combination with the data transformation layer to transform data into the model format. The mapping and configuration database can store metadata and data mappings that define the data model used by data transformation. The data and configuration user interface (UI) can facilitate access and changes to the mapping and configuration database.

[0058] In accordance with an embodiment, the data transformation layer can transform extracted data into a format suitable for loading into a customer schema of data warehouse, for example according to the data model. During the transformation, the data transformation can perform dimension generation, fact generation, and aggregate generation, as appropriate. Dimension generation can include generating dimensions or fields for loading into the data warehouse instance.

[0059] In accordance with an embodiment, after transformation of the extracted data, the data pipeline or process can execute a warehouse load procedure **150**, to load the transformed data into the customer schema of the data warehouse instance. Subsequent to the loading of the transformed data into customer schema, the transformed data can be analyzed and used in a variety of additional business intelligence processes.

[0060] Different customers of a data analytics environment may have different requirements with regard to how their data is classified, aggregated, or transformed, for purposes of providing data analytics or business intelligence data, or developing software analytic applications. In accordance with an embodiment, to support such different requirements, a semantic layer **180** can include data defining a semantic model of a customer's data; which is useful in assisting users in understanding and accessing that data using commonly-understood business terms; and provide custom content to a presentation layer **190**.

[0061] In accordance with an embodiment, a semantic model can be defined, for example, in an Oracle environment, as a BI Repository (RPD) file, having metadata that defines logical schemas, physical schemas, physical-to-logical mappings, aggregate table navigation, and/or other constructs that implement the various physical layer, business model and mapping layer, and presentation layer aspects of the semantic model.

[0062] In accordance with an embodiment, a customer may perform modifications to their data source model, to support their particular requirements, for example by adding custom facts or dimensions associated with the data stored in their data warehouse instance; and the system can extend the semantic model accordingly.

[0063] In accordance with an embodiment, the presentation layer can enable access to the data content using, for example, a software analytic application, user interface, dashboard, key performance indicators (KPI's); or other type of report or interface as may be provided by products such as, for example, Oracle Analytics Cloud, or Oracle Analytics for Applications.

[0064] In accordance with an embodiment, a query engine **18** (e.g., an OBIS instance) operates in the manner of a federated query engine to serve analytical queries or requests from clients within, e.g., an Oracle Analytics Cloud environment, directed to data stored at a database.

[0065] In accordance with an embodiment, the OBIS instance can push down operations to supported databases,

in accordance with a query execution plan **56**, wherein a logical query can include Structured Query Language (SQL) statements received from the clients; while a physical query includes database-specific statements that the query engine sends to the database to retrieve data when processing the logical query. In this way the OBIS instance translates business user queries into appropriate database-specific query languages (e.g., Oracle SQL, SQL Server SQL, DB2 SQL, or Essbase MDX). The query engine (e.g., OBIS) can also support internal execution of SQL operators that cannot be pushed down to the databases.

[0066] In accordance with an embodiment, a user/developer can interact with a client computer device **10** that includes a computer hardware **11** (e.g., processor, storage, memory), user interface **12**, and client application **14**. A query engine or business intelligence server such as OBIS generally operates to process inbound, e.g., SQL, requests against a database model, build and execute one or more physical database queries, process the data appropriately, and then return the data in response to the request.

[0067] To accomplish this, in accordance with an embodiment, the query engine or business intelligence server can include various components or features, such as a logical or business model or metadata that describes the data available as subject areas for queries; a request generator that takes incoming queries and turns them into physical queries for use with a connected data source; and a navigator that takes the incoming query, navigates the logical model and generates those physical queries that best return the data required for a particular query.

[0068] For example, in accordance with an embodiment, a query engine or business intelligence server may employ a logical model mapped to data in a data warehouse, by creating a simplified star schema business model over various data sources so that the user can query data as if it originated at a single source. The information can then be returned to the presentation layer as subject areas, according to business model layer mapping rules.

[0069] In accordance with an embodiment, the query engine (e.g., OBIS) can process queries against a database according to a query execution plan. During operation the query engine or business intelligence server can create a query execution plan which can then be further optimized, for example to perform aggregations of data necessary to respond to a request. Data can be combined together and further calculations applied, before the results are returned to the calling application.

[0070] In accordance with an embodiment, a request for data analytics or visualization information can be received via a client application and user interface as described above, and communicated to the analytics system (in the example of a cloud environment, via a cloud service). The system can retrieve an appropriate dataset to address the user/business context, for use in generating and returning the requested data analytics or visualization information to the client, as a data visualization **196**.

[0071] In accordance with an embodiment, a client application can be implemented as software or computer-readable program code executable by a computer system or processing device, and having a user interface, such as, for example, a software application user interface or a web browser interface. The client application can retrieve or access data via an Internet/HTTP or other type of network connection to

the analytics system, or in the example of a cloud environment via a cloud service provided by the environment.

[0072] FIG. 2 further illustrates an example data analytics environment, in accordance with an embodiment.

[0073] As illustrated in FIG. 2, in accordance with an embodiment, the analytics system enables a dataset to be retrieved, received, or prepared from one or more data source(s) **198**, for example via one or more data source connections. Examples of the types of data that can be transformed, analyzed, or visualized using the systems and methods described herein include HCM, HR, or ERP data, e-mail or text messages, or other of free-form or unstructured textual data provided at one or more of a database, data storage service, or other type of data repository or data source.

[0074] For example, in accordance with an embodiment, a request for data analytics or visualization information can be received via a client application and user interface as described above, and communicated to the analytics system (in the example of a cloud environment, via a cloud service). The system can retrieve an appropriate dataset to address the user/business context, for use in generating and returning the requested data analytics or visualization information to the client. For example, the data analytics system can retrieve a dataset using, e.g., SELECT statements or Logical SQL instructions.

[0075] In accordance with an embodiment, the system provides functionality that allows a user to generate datasets, analyses, or visualizations for display within a user interface, for example to explore datasets or data sourced from multiple data sources.

[0076] In accordance with an embodiment, the provisioning component can also comprise a provisioning application programming interface (API), a number of workers, a metering manager, and a data plane API, as further described below. The console interface can communicate, for example, by making API calls, with the provisioning API when commands, instructions, or other inputs are received at the console interface to provision services within the SaaS environment, or to make configuration changes to provisioned services.

[0077] In accordance with an embodiment, the data plane API can communicate with the data plane. For example, in accordance with an embodiment, provisioning and configuration changes directed to services provided by the data plane can be communicated to the data plane via the data plane API.

[0078] In accordance with an embodiment, the metering manager can include various functionality that meters services and usage of services provisioned through control plane. For example, in accordance with an embodiment, the metering manager can record a usage over time of processors provisioned via the control plane, for particular customers (tenants), for billing purposes. Likewise, the metering manager can record an amount of storage space of data warehouse partitioned for use by a customer of the SaaS environment, for billing purposes.

[0079] In accordance with an embodiment, the data pipeline or process, provided by the data plane, can include a monitoring component, a data staging component, a data quality component, and a data projection component, as further described below.

[0080] In accordance with an embodiment, the data transformation layer can include a dimension generation compo-

nent, fact generation component, and aggregate generation component, as further described below. The data plane can also include a data and configuration user interface, and mapping and configuration database.

[0081] In accordance with an embodiment, the data warehouse can include a default data analytics schema (referred to herein in accordance with some embodiments as an analytic warehouse schema) **162** and, for each customer (tenant) of the system, a customer schema **164**.

[0082] In accordance with an embodiment, to support multiple tenants, the system can enable the use of multiple data warehouses or data warehouse instances. For example, in accordance with an embodiment, a first warehouse customer tenancy for a first tenant can comprise a first database instance, a first staging area, and a first data warehouse instance of a plurality of data warehouses or data warehouse instances; while a second customer tenancy for a second tenant can comprise a second database instance, a second staging area, and a second data warehouse instance of the plurality of data warehouses or data warehouse instances.

[0083] In accordance with an embodiment, based on the data model defined in the mapping and configuration database, the monitoring component can determine dependencies of several different datasets (data sets) to be transformed. Based on the determined dependencies, the monitoring component can determine which of several different datasets should be transformed to the model format first.

[0084] For example, in accordance with an embodiment, if a first model dataset includes no dependencies on any other model dataset; and a second model dataset includes dependencies to the first model dataset; then the monitoring component can determine to transform the first dataset before the second dataset, to accommodate the second dataset's dependencies on the first dataset.

[0085] For example, in accordance with an embodiment, dimensions can include categories of data such as, for example, "name," "address," or "age". Fact generation includes the generation of values that data can take, or "measures." Facts can be associated with appropriate dimensions in the data warehouse instance. Aggregate generation includes creation of data mappings which compute aggregations of the transformed data to existing data in the customer schema of data warehouse instance.

[0086] In accordance with an embodiment, once any transformations are in place (as defined by the data model), the data pipeline or process can read the source data, apply the transformation, and then push the data to the data warehouse instance.

[0087] In accordance with an embodiment, data transformations can be expressed in rules, and once the transformations take place, values can be held intermediately at the staging area, where the data quality component and data projection components can verify and check the integrity of the transformed data, prior to the data being uploaded to the customer schema at the data warehouse instance. Monitoring can be provided as the extract, transform, load process runs, for example, at a number of compute instances or virtual machines. Dependencies can also be maintained during the extract, transform, load process, and the data pipeline or process can attend to such ordering decisions.

[0088] In accordance with an embodiment, after transformation of the extracted data, the data pipeline or process can execute a warehouse load procedure, to load the transformed

data into the customer schema of the data warehouse instance. Subsequent to the loading of the transformed data into customer schema, the transformed data can be analyzed and used in a variety of additional business intelligence processes.

[0089] FIG. 3 further illustrates an example data analytics environment, in accordance with an embodiment.

[0090] As illustrated in FIG. 3, in accordance with an embodiment, data can be sourced, e.g., from a customer's (tenant's) enterprise software application or data environment (**106**), using the data pipeline process; or as custom data **109** sourced from one or more customer-specific applications **107**; and loaded to a data warehouse instance, including in some examples the use of an object storage **105** for storage of the data.

[0091] In accordance with embodiments of analytics environments such as, for example, Oracle Analytics Cloud (OAC), a user can create a dataset that uses tables from different connections and schemas. The system uses the relationships defined between these tables to create relationships or joins in the dataset.

[0092] In accordance with an embodiment, for each customer (tenant), the system uses the data analytics schema that is maintained and updated by the system, within a system/cloud tenancy **114**, to pre-populate a data warehouse instance for the customer, based on an analysis of the data within that customer's enterprise applications environment, and within a customer tenancy **117**. As such, the data analytics schema maintained by the system enables data to be retrieved, by the data pipeline or process, from the customer's environment, and loaded to the customer's data warehouse instance.

[0093] In accordance with an embodiment, the system also provides, for each customer of the environment, a customer schema that is readily modifiable by the customer, and which allows the customer to supplement and utilize the data within their own data warehouse instance. For each customer, their resultant data warehouse instance operates as a database whose contents are partly-controlled by the customer; and partly-controlled by the environment (system).

[0094] For example, in accordance with an embodiment, a data warehouse (e.g., ADW) can include a data analytics schema and, for each customer/tenant, a customer schema sourced from their enterprise software application or data environment. The data provisioned in a data warehouse tenancy (e.g., an ADW cloud tenancy) is accessible only to that tenant; while at the same time allowing access to various, e.g., ETL-related or other features of the shared environment.

[0095] In accordance with an embodiment, to support multiple customers/tenants, the system enables the use of multiple data warehouse instances; wherein for example, a first customer tenancy can comprise a first database instance, a first staging area, and a first data warehouse instance; and a second customer tenancy can comprise a second database instance, a second staging area, and a second data warehouse instance.

[0096] In accordance with an embodiment, for a particular customer/tenant, upon extraction of their data, the data pipeline or process can insert the extracted data into a data staging area for the tenant, which can act as a temporary staging area for the extracted data. A data quality component and data protection component can be used to ensure the integrity of the extracted data; for example by performing

validations on the extracted data while the data is temporarily held in the data staging area. When the extract process has completed its extraction, the data transformation layer can be used to begin the transformation process, to transform the extracted data into a model format to be loaded into the customer schema of the data warehouse.

[0097] FIG. 4 further illustrates an example data analytics environment, in accordance with an embodiment.

[0098] As illustrated in FIG. 4, in accordance with an embodiment, the process of extracting data, e.g., from a customer's (tenant's) enterprise software application or data environment, using the data pipeline process as described above; or as custom data sourced from one or more customer-specific applications; and loading the data to a data warehouse instance, or refreshing the data in a data warehouse, generally involves three broad stages, performed by an ETP service 160 or process, including one or more extraction service 163; transformation service 165; and load/publish service 167, executed by one or more compute instance(s) 170.

[0099] For example, in accordance with an embodiment, a list of view objects for extractions can be submitted, for example, to an Oracle BI Cloud Connector (BICC) component via a REST call. The extracted files can be uploaded to an object storage component, such as, for example, an Oracle Storage Service (OSS) component, for storage of the data. The transformation process takes the data files from object storage component (e.g., OSS), and applies a business logic while loading them to a target data warehouse, e.g., an ADW database, which is internal to the data pipeline or process, and is not exposed to the customer (tenant). A load/publish service or process takes the data from the, e.g., ADW database or warehouse, and publishes it to a data warehouse instance that is accessible to the customer (tenant).

[0100] FIG. 5 further illustrates an example data analytics environment, in accordance with an embodiment.

[0101] As illustrated in FIG. 5, which illustrates the operation of the system with a plurality of tenants (customers) in accordance with an embodiment, data can be sourced, e.g., from each of a plurality of customer's (tenant's) enterprise software application or data environment, using the data pipeline process as described above; and loaded to a data warehouse instance.

[0102] In accordance with an embodiment, the data pipeline or process maintains, for each of a plurality of customers (tenants), for example customer A 180, customer B 182, a data analytics schema that is updated on a periodic basis, by the system in accordance with best practices for a particular analytics use case.

[0103] In accordance with an embodiment, for each of a plurality of customers (e.g., customers A, B), the system uses the data analytics schema 162A, 162B, that is maintained and updated by the system, to pre-populate a data warehouse instance for the customer, based on an analysis of the data within that customer's enterprise applications environment 106A, 106B, and within each customer's tenancy (e.g., customer A tenancy 181, customer B tenancy 183); so that data is retrieved, by the data pipeline or process, from the customer's environment, and loaded to the customer's data warehouse instance 160A, 160B.

[0104] In accordance with an embodiment, the data analytics environment also provides, for each of a plurality of customers of the environment, a customer schema (e.g.,

customer A schema 164A, customer B schema 164B) that is readily modifiable by the customer, and which allows the customer to supplement and utilize the data within their own data warehouse instance.

[0105] As described above, in accordance with an embodiment, for each of a plurality of customers of the data analytics environment, their resultant data warehouse instance operates as a database whose contents are partly-controlled by the customer; and partly-controlled by the data analytics environment (system); including that their database appears pre-populated with appropriate data that has been retrieved from their enterprise applications environment to address various analytics use cases. When the extract process 108A, 108B for a particular customer has completed its extraction, the data transformation layer can be used to begin the transformation process, to transform the extracted data into a model format to be loaded into the customer schema of the data warehouse.

[0106] In accordance with an embodiment, activation plans 186 can be used to control the operation of the data pipeline or process services for a customer, for a particular functional area, to address that customer's (tenant's) particular needs.

[0107] For example, in accordance with an embodiment, an activation plan can define a number of extract, transform, and load (publish) services or steps to be run in a certain order, at a certain time of day, and within a certain window of time.

[0108] In accordance with an embodiment, each customer can be associated with their own activation plan(s). For example, an activation plan for a first Customer A can determine the tables to be retrieved from that customer's enterprise software application environment (e.g., their Fusion Applications environment), or determine how the services and their processes are to run in a sequence; while an activation plan for a second Customer B can likewise determine the tables to be retrieved from that customer's enterprise software application environment, or determine how the services and their processes are to run in a sequence.

[0109] FIG. 6 illustrates a use of the system to transform, analyze, or visualize data, in accordance with an embodiment.

[0110] As illustrated in FIG. 6, in accordance with an embodiment, the systems and methods disclosed herein can be used to provide a data visualization environment 192 that enables insights for users of an analytics environment with regard to analytic artifacts and relationships among the same. A model can then be used to visualize relationships between such analytic artifacts via, e.g., a user interface, as a network chart or visualization of relationships and lineage between artifacts (e.g., User, Role, DV Project, Dataset, Connection, Dataflow, Sequence, ML Model, ML Script).

[0111] In accordance with an embodiment, a client application can be implemented as software or computer-readable program code executable by a computer system or processing device, and having a user interface, such as, for example, a software application user interface or a web browser interface. The client application can retrieve or access data via an Internet/HTTP or other type of network connection to the analytics system, or in the example of a cloud environment via a cloud service provided by the environment.

[0112] In accordance with an embodiment, the user interface can include or provide access to various dataflow action types, as described in further detail below, that enable

self-service text analytics, including allowing a user to display a dataset, or interact with the user interface to transform, analyze, or visualize the data, for example to generate graphs, charts, or other types of data analytics or visualizations of dataflows.

[0113] In accordance with an embodiment, the analytics system enables a dataset to be retrieved, received, or prepared from one or more data source(s), for example via one or more data source connections. Examples of the types of data that can be transformed, analyzed, or visualized using the systems and methods described herein include HCM, HR, or ERP data, e-mail or text messages, or other of free-form or unstructured textual data provided at one or more of a database, data storage service, or other type of data repository or data source.

[0114] For example, in accordance with an embodiment, a request for data analytics or visualization information can be received via a client application and user interface as described above, and communicated to the analytics system (in the example of a cloud environment, via a cloud service). The system can retrieve an appropriate dataset to address the user/business context, for use in generating and returning the requested data analytics or visualization information to the client. For example, the data analytics system can retrieve a dataset using, e.g., SELECT statements or Logical SQL instructions.

[0115] In accordance with an embodiment, the system can create a model or dataflow that reflects an understanding of the dataflow or set of input data, by applying various algorithmic processes, to generate visualizations or other types of useful information associated with the data. The model or dataflow can be further modified within a dataset editor **193** by applying various processing or techniques to the dataflow or set of input data, including for example one or more dataflow actions **194**, **195** or steps that operate on the dataflow or set of input data. A user can interact with the system via a user interface, to control the use of dataflow actions to generate data analytics, data visualizations **196**, or other types of useful information associated with the data.

[0116] In accordance with an embodiment, datasets are self-service data models that a user can build for data visualization and analysis requirements. A dataset contains data source connection information, tables, and columns, data enrichments and transformations. A user can use a dataset in multiple workbooks and in dataflows.

[0117] In accordance with an embodiment, when a user creates and builds a dataset, they can, for example: choose between many types of connections or spreadsheets; create datasets based on data from multiple tables in a database connection, an Oracle data source, or a local subject area; or create datasets based on data from tables in different connections and subject areas.

[0118] For example, in accordance with an embodiment, a user can build a dataset that includes tables from an Autonomous Data Warehouse connection, tables from a Spark connection, and tables from a local subject area; specify joins between tables; and transform and enrich the columns in the dataset.

[0119] In accordance with an embodiment, additional artifacts, features, and operations associated with datasets can include, for example:

[0120] View available connections: a dataset uses one or more connections to data sources to access and supply data for analysis and visualization. A user list of connections

contains the connections that they built and the connections that they have permission to access and use.

[0121] Create a dataset from a connection: when a user creates a dataset, they can add tables from one or more data source connections, add joins, and enrich data.

[0122] Add multiple connections to a dataset: a dataset can include more than one connection. Adding more connections allows a user to access and join all of the tables and data that they need to build the dataset. The user can add more connections to datasets that support multiple tables.

[0123] Create dataset table joins: joins indicate relationships between a dataset's tables. If the user is creating a dataset based on facts and dimensions and if joins already exist in the source tables, then joins are automatically created in the dataset. If the user is creating a dataset from multiple connections and schemas, then they can manually define the joins between tables.

[0124] In accordance with an embodiment, a user can use dataflows to create datasets by combining, organizing, and integrating data. Dataflows enable the user to organize and integrate data to produce curated datasets that either they or other users can visualize.

[0125] For example, in accordance with an embodiment, a user might use a dataflow to: Create a dataset; Combine data from different source; aggregate data; and train a machine learning model or apply a predictive machine learning model to their data.

[0126] In accordance with an embodiment, a dataset editor as described above allows a user to add actions or steps, wherein each step performs a specific function, for example, add data, join tables, merge columns, transform data, or save the data. Each step is validated when the user adds or changes it. When they have configured the dataflow, they can execute it to produce or update a dataset.

[0127] In accordance with an embodiment, a user can curate data from datasets, subject areas, or database connections. The user can execute dataflows individually or in a sequence. The user can include multiple data sources in a dataflow and specify how to join them. The user can save the output data from a dataflow in either a dataset or in a supported database type.

[0128] In accordance with an embodiment, additional artifacts, features, and operations associated with dataflows can include, for example:

[0129] Add columns: add custom columns to a target dataset.

[0130] Add data: add data sources to a dataflow. For example, if the user is merging two datasets, they add both datasets to the dataflow.

[0131] Aggregate: create group totals by applying aggregate functions; for example, count, sum, or average.

[0132] Branch: creates multiple outputs from a dataflow.

[0133] Filter: select only the data that the user is interested in.

[0134] Join: combine data from multiple data sources using a database join based on a common column.

[0135] Graph Analytics: perform geo-spatial analysis, such as calculating the distance or the number of hops between two vertices.

[0136] The above are provided by way of example; in accordance with an embodiment other types of steps can be added to a dataflow to transform a dataset or provide data analytics or visualizations.

Dataset Analyses and Visualizations

[0137] FIGS. 7-8 illustrate the preparation of a data visualization for use with a data analytics environment, in accordance with an embodiment.

[0138] As illustrated in FIGS. 7-8, in accordance with an embodiment, the system allows a user to prepare a data visualization for use with a dataset. For example, a panel of visualization options allows users to add data visualization elements to a workbook or canvas, to create a dashboard or data visualization.

[0139] For example, in accordance with an embodiment, the user can create a workbook, add a dataset, and then drag and drop its columns onto a canvas to create visualizations.

[0140] In accordance with an embodiment, the user can choose between various types of connections to create datasets based on data from tables in, e.g., a database connection, an Oracle subject area, an Oracle ADW connection, or a spreadsheet, file, or other type of data source.

[0141] In accordance with an embodiment, the system can automatically generate a visualization based on the contents of the canvas, with one or more visualization types automatically selected for selection by the user. For example, if the user adds a revenue measure to the canvas, the data element may be placed in a values area of a grammar panel, and a Tile visualization type selected.

[0142] In accordance with an embodiment, the system can provide automatically generated data visualizations (automatically-generated insights, auto-insights), by suggesting visualizations which are expected to provide the best insights for a particular dataset.

[0143] FIG. 9 further illustrates the preparation of a data visualization for use with a data analytics environment, in accordance with an embodiment.

[0144] As illustrated in FIG. 9, in accordance with an embodiment, the user can continue adding data elements directly to the canvas to build the visualization. In this manner, a dataset operates as a self-service data model from which the user can build a data analysis or visualization. The user can then use the data visualization to access the data analytics environment, for example to submit analyses or queries against an organization's data.

[0145] The user interfaces and features shown in FIGS. 7-9 and elsewhere herein are provided by way of example, for purposes of illustration of the various features described herein; in accordance with various embodiments, alternative examples of user interfaces and features can be provided.

[0146] In accordance with an embodiment, the system provides functionality that allows a user to generate datasets, analyses, or visualizations for display within a user interface, for example to explore datasets or data sourced from multiple data sources.

[0147] For example, in accordance with an embodiment, the user can choose between various types of connections to create datasets based on data from tables in, e.g., a database connection, an Oracle subject area, an Oracle ADW connection, or a spreadsheet, file, or other type of data source. In this manner, a dataset operates as a self-service data model from which the user can build a data analysis or visualization.

[0148] In accordance with an embodiment, a dataset editor can display a list of connections which the user has permission to access, and allow the user to create or edit a dataset that includes tables, joins, and/or enriched data. The editor can display the data source connection's schemas and tables,

from which the user can drag and drop to a dataset diagram. If a particular connection does not itself provide a schema and table listing the user can use a manual query for appropriate tables. Adding connections provide the ability to access and join their associated tables and data, to build the dataset.

[0149] In accordance with an embodiment, within the dataset editor a join diagram displays the tables and joins in a dataset. Joins that are defined in the data source can be automatically created between tables in the dataset, for example, by creating joins based on column name matches found between the tables.

[0150] In accordance with an embodiment, when the user selects a table, a preview data area displays a sample of the table's data. Displayed join links and icons indicate which tables are joined and the type of join used. The user can create a join by dragging and dropping one table onto another; click on a join to view or update its configuration; or click a column's type attribute to change its type, for example from a measure to an attribute.

[0151] In accordance with an embodiment, the system can generate source-specific optimized queries for a visualization, wherein a dataset is treated as a data model and only those tables needed to satisfy a visualization are used in the query.

[0152] By default a dataset's grain is determined by the table with the lowest grain. The user can create a measure in any table in a dataset; however, this can cause the measure on one side of a one-to-many or many-to-many relationship to be duplicated. In accordance with an embodiment, to address this, the user can set the table on one side of a cardinality to preserve grain, to keep its level of detail.

[0153] In accordance with an embodiment, dataset tables can be associated with a data access setting that determines if the system will load the table into cache, or alternatively if the table will receive its data directly from the data source.

[0154] In accordance with an embodiment, when automatic caching mode is selected for a table, the system loads or reloads the table data into cache, which provides faster performance when the table's data is refreshed, e.g., from a workbook, and causes the reload menu option to display at the table and dataset level.

[0155] In accordance with an embodiment, when live mode is selected for a table, the system retrieves the table data directly from the data source; and the source system manages the table's data source queries. This option is useful when the data is stored in a high-performance data warehouse such as, for example, Oracle ADW; and also ensures that the most-current data is used.

[0156] In accordance with an embodiment, when a dataset uses multiple tables, some tables can use automatic caching, while others can include live data. During reload of multiple tables using the same connection, if the reloading of data on one table fails, then any tables presently set to use automatic caching are switched to using live mode to retrieve their data.

[0157] In accordance with an embodiment, the system allows a user to enrich and transform their data before it is made available for analysis. When a workbook is created and a dataset added to it, the system performs column level profiling on a representative sample of the data. After profiling the data, the user can implement transformation and enrichment recommendations provided for recognizable

columns in the dataset; such as, for example, GPS enrichments such as latitude and longitude for cities or zip codes.

[0158] In accordance with an embodiment, the data transformation and enrichment changes applied to a dataset affect the workbooks and dataflows that use the dataset. For example, when the user opens a workbook that shares the dataset, they receive a message indicating that the workbook uses updated or refreshed data.

[0159] In accordance with an embodiment, dataflows provide a means of organizing and integrating data to produce curated datasets that your users can visualize. For example, the user might use a dataflow to create a dataset, combine data from different sources, aggregate data, or train machine learning models or apply a predictive machine learning model to their data.

[0160] In accordance with an embodiment, within a dataflow each step performs a specific function, for example to add data, join tables, merge columns, transform data, or save data. Once configured, the dataflow can be executed to perform operations to produce or update a dataset, including for example the use of SQL operators, such as BETWEEN, LIKE, IN), conditional expressions, or functions.

[0161] In accordance with an embodiment, dataflows can be use merge datasets, cleanse data, and output the results to a new dataset. Dataflows can be executed individually or in a sequence. If any dataflow within a sequence fails, then all the changes made in the sequence are rolled back.

[0162] In accordance with an embodiment, visualizations can be displayed within a user interface, for example to explore datasets or data sourced from multiple data sources, and to add insights.

[0163] For example, in accordance with an embodiment, the user can create a workbook, add a dataset, and then drag and drop its columns onto a canvas to create visualizations. The system can automatically generate a visualization based on the contents of the canvas, with one or more visualization types automatically selected for selection by the user. For example, if the user adds a revenue measure to the canvas, the data element may be placed in a values area of a grammar panel, and a Tile visualization type selected. The user can continue adding data elements directly to the canvas to build the visualization.

[0164] In accordance with an embodiment, the system can provide automatically generated data visualizations (automatically-generated insights, auto-insights), by suggesting visualizations which are expected to provide the best insights for a particular dataset. The user can review an insight's automatically generated summary, for example by hovering over the associated visualization in the workbook canvas.

Data Analytics Assistant

[0165] In accordance with an embodiment, described herein is a system and method for providing a data analytics assistant (data analytics workbook assistant), for use with a data analytics environment. In accordance with an embodiment a data analytics system or environment can be integrated with a digital assistant which provides natural language processing capabilities, for purposes of leveraging the natural language processing of a user's text or speech input, within a data analytics or data visualization project, for example while generating, modifying, or interacting with data visualizations, or generating a story or script that includes or is descriptive of data visualizations.

[0166] For example, in accordance with an embodiment a data analytics system or environment, for example an Oracle Analytics Cloud (OAC) environment, can be integrated with a digital assistant system or environment, for example an Oracle Digital Assistant (ODA) environment, which provides natural language processing (NLP) and speech processing capabilities, for purposes of leveraging the natural language (NL) processing of a user's text or speech input, within a data analytics or data visualization project, for example while generating, modifying, or interacting with data visualizations.

[0167] FIG. 10 illustrates a system for providing digital assistant integration with a data analytics assistant, in accordance with an embodiment.

[0168] As illustrated in FIG. 10, in accordance with an embodiment, at (1) a data analytics system or environment, for example an Oracle Analytics Cloud (OAC) environment, receives as input from a user via a user interface (e.g., data analytics assistant) a natural language expression, or request to prepare a data visualization.

[0169] At (2), the input natural language can be associated with a context where appropriate, for example an instruction to create a project, e.g., visualization, story, script.

[0170] At (3), the natural language expression is sent to a digital assistant system or environment, for example an Oracle Digital Assistant (ODA) environment, which provides natural language processing (NLP) and speech processing capabilities, for purposes of leveraging the natural language (NL) processing of the natural language expression, e.g., a user's text or speech input.

[0171] At (4), an (e.g., ODA) JSON data is prepared with resolved intent and entities.

[0172] At (5), a search component, e.g., OBIS, BISearch, finds a most relevant dataset based on the parsed JSON data, and accessibility for the user, and generates a project JSON and returns to the data visualization (DV) environment for rendering.

[0173] At (6), the data analytics or data visualization project is rendered in the user interface (UI).

[0174] In accordance with an embodiment, a natural language generator (NLG) service within OAC can generate simple and insightful natural language text for a given visualization. A simple text explains the data behind the visualization, whereas an insightful text is meant to provide related but useful insights about the columns and the data surrounding them in the visualization. The data analytics assistant can then use the insights text generation feature to fetch and display related insights for the visualization.

[0175] FIG. 11 illustrates the use of a natural language generator service to support digital assistant integration, in accordance with an embodiment.

[0176] In accordance with an embodiment, as used with the overall data analytics assistant feature, the NLG service can include a data collector responsible for generating the insightful data for a given visualization. To accomplish this, it takes the visualization metadata as input. The metadata is processed to extract an input grammar for the NLG service. For example, the input grammar is made up of the projections, group by and filter expressions, dimension and measure columns and any other aspects of the visualization that can be of potential use in generating insights data.

[0177] In accordance with an embodiment, the input grammar is pruned using the dataset profile to generate insights grammar. The process of pruning applies transforms

to generate insights grammar. For example, one of the transformations is to determine a dimension column either from the input grammar or the dataset to explain the measure in the input visualization. Based on this, a rank filter predicate is added to the grammar. This is just one example of the transformation that aids in fetching a top N insights.

[0178] In accordance with an embodiment, the final step is to convert the insights grammar into a logical SQL that can be executed to fetch insights data.

Native Visualization Tool

[0179] In accordance with an embodiment, the system can include a real-time speech recognition and native visualization tool, for use with data analytics environments. The described approach can be used to provide real-time data visualizations or other analytics based on a live conversation with a user using an on-device speech-to-text recognition engine.

[0180] In accordance with an embodiment, the described approach alleviates the need for a technical expert versed in the use of an analytics tool or software to build charts and insights, by allowing the user to have a natural conversation with the device that can catch relevant mentions about the data, and turn them into graphical representations in a real-time and immediate manner.

[0181] For example, in accordance with an embodiment, the user can customize a visualization provided in real-time by editing columns, filtering data, changing chart type, sorting data, sharing it as a picture to other applications (apps), adding it to a watchlist, or creating a new workbook; all achieved through their voice inputs.

[0182] FIGS. 12A-12B illustrate the use of a tool for providing real-time speech recognition and native visualization for data analytics, in accordance with an embodiment.

[0183] As illustrated in FIGS. 12A-12B, in accordance with an embodiment, the system comprises a client mobile application (app) that operates in combination with a microphone 202, and on-device speech-to-text engine 204, to communicate with a server API 212 (for example, a data analytics assistant API) and mobile API 214, and allow a user to interact with the mobile application and to receive, from a data analytics environment, real-time data visualizations or other analytics information.

[0184] In accordance with an embodiment, the system operates to receive partial results from a speech recognition model built-in to the device (for example, in Android or iOS device environment as can be provided, respectively, by an Android or iOS SDK; other types of devices can include other types of speech-to-text engines or speech recognition models).

[0185] In accordance with an embodiment, the partial results are transformed into individual utterances by processing them inside the mobile application, and detecting splits to transform them into sentences. Completed sentences can then be sent to a data analytics environment service via a server API 212 and/or mobile API, that can then either return relevant results, or nothing to process.

[0186] In accordance with an embodiment, the mobile application can operate in the manner of a state engine to ignore those results that are not relevant to its current state or context, while for relevant responses display those natively as a graphical representation, for example as a data visualization or other analytics information.

[0187] In accordance with an embodiment, the system can enable the use of one or more languages supported by the on-device speech-to-text engine, for purposes of allowing a user to interact with the mobile application to receive, from the data analytics environment, real-time data visualizations or other analytics information. The described approach allows the mobile application to natively render visualizations in a fast and pixel-perfect style, in contrast to other systems that may require the use of embedded tools to display results. Additionally, unlike systems that may require extensive training of a voice model, the described approach does not require such training, and instead voice commands can be sent, and the server can provide relevant results in real-time, with no extra effort required by the user to set up.

[0188] FIG. 13 illustrates a process for providing real-time speech recognition and native visualization for data analytics, in accordance with an embodiment.

[0189] As illustrated in FIG. 13, in accordance with an embodiment, partial results are transformed into individual utterances by processing them inside the mobile application, and detecting splits to transform them into sentences. Completed sentences can then be sent to a data analytics environment service via a server API that can then either return relevant results, or operate with nothing to process at the present time.

[0190] For example, as illustrated in FIG. 13 by way of example, in accordance with an embodiment, upon starting the feature, a dataset is retrieved according to a dataset ID. A determination is made as to whether use of the microphone is allowed. If not, then permission is requested. A data analytics assistant mobile view is then started, and bias words are set in the speech-to-text-engine in the mobile device. The setting of such bias or context within the speech-to-text-engine for particular words helps improve the detection of specific content—for example known prompts or dataset column labels.

[0191] In accordance with an embodiment, the mobile device will then listen for audio (words) received as input from the user. As the user provides input, the device operates to detect pauses for sentence consideration, and build an array of words associated with the sentence. If the sentence is determined to be a mobile custom prompt, i.e., capable of being processed without additional server data (for example, add to watchlist, add to workbook, or a variation of roll-back) then the device can perform the step locally, without contacting the server. Alternatively, the device will send the input sentence to the server API, in order to determine an analytics assistant intent. Such intent may depend on the mobile device state; in some instances the server might return an empty prompt.

[0192] In accordance with an embodiment, the process concludes, as described above, with the server returning relevant results, or operating with nothing to process at the present time.

[0193] FIG. 14 further illustrates a process for providing real-time speech recognition and native visualization for data analytics, in accordance with an embodiment.

[0194] As illustrated in FIG. 14, in accordance with an embodiment, for relevant results that indicate a new data visualization, the mobile application can directly execute a request to the mobile API inside the data analytics environment, that will provide in table and raw form the data needed for this visualization.

[0195] For example, as illustrated in FIG. 14 by way of example, in accordance with an embodiment, upon activating the mobile device microphone, audio (words) received as input from the user are processed by the on-device speech-to-text engine. As the user provides input, the device associated words with timestamps and sends those to the mobile application which operates to detect pauses for sentence consideration, and build an array of words associated with the sentence. If the sentence is determined to be an on-device prompt, then the device can perform the step locally, without contacting the server. Otherwise, the server API can determine an intent associated with audio (words) received as input, which is sent to the mobile application, which then communicates with the mobile API, for example to request data for a new visualization.

[0196] In accordance with an embodiment, a similar approach can be used to determine if a sentence input to, for example, update an existing visualization, can be validated and performed on-device, without contacting the server. Results that are not relevant to a current state or context can be ignored.

[0197] In accordance with an embodiment, a long input pause can be determined by the mobile application to indicate the end of the users' session. The mobile application can then, for example, display a final result, and close the listening cycle.

[0198] FIGS. 15A-15S illustrate an example use of a tool for providing real-time speech recognition and native visualization for data analytics, in accordance with an embodiment.

[0199] For example, as illustrated in FIGS. 15A-15S, in accordance with an embodiment, a user can open the mobile application on their device and specify a dataset to work with. Subsequent utterances from the user are then examined in the context of that dataset.

[0200] In accordance with an embodiment, the system operates in an always-listening mode to receive partial results of the input voice. Each partial result is sent to the server API, which ignores prompts not related to the data, or waits until there is sufficient input to process.

[0201] In accordance with an embodiment, utterances received are aligned with a time stamp, each of which make a partial result, such that a complete sentence may include multiple partial results. Once the server API (for example, a data analytics assistant API) has something that it understands, it generates a response. Since the mobile application can operate in the manner of a state engine, it can ignore those results that are not relevant to its current state or context. Otherwise, an intent is sent to the mobile application, which can operate accordingly, for example to communicate with the mobile API to request data for a new visualization.

[0202] For example, in accordance with an embodiment illustrated in FIGS. 15A-15S, in response to a user voice input to "Show me profit by segment", the mobile application can communicate with the mobile API to receive, from the data analytics environment, an appropriate data, for purpose of generating the data visualization.

[0203] In accordance with an embodiment, for operations within the mobile application directed to sorting (e.g., a user voice input to "Sort ascending on profit") or changing a chart type (e.g., a user voice input to "Change chart type to line") the mobile application does not need to go back to the server

API (for example, a data analytics assistant API)—since this operation can be performed completely on-device.

[0204] Similarly in accordance with an embodiment, on-device prompts to, e.g., "roll-back" to displaying a prior visualization, are handled within the mobile application, without going to the server.

Technical Advantages

[0205] In accordance with an embodiment, various advantages of the described approach include, for example:

[0206] Increase platform usage by integrating a mobile application real-time speech-to-text feature as an option to consume data analytics information or visualizations.

[0207] Immerse data analytics into artificial intelligence fields by using audio and visualization generation tools to consume analytics.

[0208] Leverage data analytics information or visualizations by bringing it to a mobile platform accessible to any end user with voice and native visualizations. This also improves productivity for customers and can be applied to numerous use cases where a mobile application can be used, such as meetings, webinars, personal analytics, financial reports.

[0209] Facilitate the creation of charts, insights or any other type of visualization without a technical skill and in a natural way of speaking to a mobile device.

[0210] Bring smart and sophisticated features to the platform that can detect and deliver immediate results easily.

[0211] Cost savings in speech recognition engine by using built-in solutions in mobile phone and process them locally to get immediate results.

[0212] Data security by handling results locally, since what users will be speaking and the mobile application detects, will remain secure and won't travel to other services.

[0213] The tool is designed to scale effortlessly, making it suitable to any size of businesses or enterprises. Extending the speech detections and enabling customizations for provided visualizations will be easy to implement.

[0214] In accordance with various embodiments, the teachings herein can be implemented using one or more computer, computing device, machine, or microprocessor, including one or more processors, memory and/or computer readable storage media programmed according to the teachings herein. Appropriate software coding can readily be prepared by skilled programmers based on the teachings of the present disclosure, as will be apparent to those skilled in the software art.

[0215] In some embodiments, the teachings herein can include a computer program product which is a non-transitory computer readable storage medium (media) having instructions stored thereon/in which can be used to program a computer to perform any of the processes of the present teachings. Examples of such storage mediums can include, but are not limited to, hard disk drives, hard disks, hard drives, fixed disks, ROMs, RAMs, EPROMs, EEPROMs, DRAMs, VRAMs, flash memory devices, or other types of storage media or devices suitable for non-transitory storage of instructions and/or data.

[0216] The foregoing description has been provided for the purposes of illustration and description. It is not intended to be exhaustive or to limit the scope of protection to the precise forms disclosed. Further modifications and variations will be apparent to the practitioner skilled in the art.

[0217] For example, although several of the examples provided herein illustrate operation with an enterprise software application or data analytics environment such as, for example, an Oracle Analytics Cloud environment; in accordance with various embodiments, the systems and methods described herein can be used with other types of enterprise software application or data environments, cloud environments, cloud services, cloud computing, or other computing environments.

[0218] The embodiments were chosen and described in order to best explain the principles of the teachings herein and their practical application, thereby enabling others skilled in the art to understand the various embodiments and with various modifications that are suited to the particular use contemplated. It is intended that the scope be defined by the following claims and their equivalents.

What is claimed is:

1. A system for providing a real-time speech recognition and native visualization tool, for use with data analytics environments, comprising:

an on-device mobile application that operates in combination with an on-device speech-to-text engine to communicate with a data analytics system or environment for purposes of generating, modifying, or interacting with data visualizations, including:

listening for audio received as input from a user;

as the user provides input, detecting pauses for sentence consideration, and building an array of words associated with the sentence;

determining whether the sentence is capable of being processed without additional server data and if so performing the step locally; and

alternatively sending an input sentence to a server API, in order to determine an intent associated therewith for use by the server in returning relevant results.

2. The system of claim 1, wherein the data analytics system or environment is an analytics cloud environment.

3. The system of claim 1, wherein the data analytics system or environment includes a data analytics assistant provided by or for use with the data analytics environment in generating data visualizations associated with datasets.

4. The system of claim 1, wherein:

the data analytics system or environment receives as input a natural language expression, and the input natural language expression is associated with a context;

a search component finds a most relevant dataset, which is returned to a data visualization environment for rendering within a user interface.

5. The system of claim 1, wherein the data analytics system or environment and digital assistant system or environment are provided or communicate as part of a cloud environment.

6. A method for providing a real-time speech recognition and native visualization tool, for use with data analytics environments, comprising:

providing an on-device mobile application that operates in combination with an on-device speech-to-text engine to communicate with a data analytics system or environment for purposes of generating, modifying, or interacting with data visualizations, including:

listening for audio received as input from a user;

as the user provides input, detecting pauses for sentence consideration, and building an array of words associated with the sentence;

determining whether the sentence is capable of being processed without additional server data and if so performing the step locally; and

alternatively sending an input sentence to a server API, in order to determine an intent associated therewith for use by the server in returning relevant results.

7. The method of claim 6, wherein the data analytics system or environment is an analytics cloud environment.

8. The method of claim 6, wherein the data analytics system or environment includes a data analytics assistant provided by or for use with the data analytics environment in generating data visualizations associated with datasets.

9. The method of claim 6, wherein:

the data analytics system or environment receives as input a natural language expression, and the input natural language expression is associated with a context;

a search component finds a most relevant dataset, which is returned to a data visualization environment for rendering within a user interface.

10. The method of claim 6, wherein the data analytics system or environment and digital assistant system or environment are provided or communicate as part of a cloud environment.

11. A non-transitory computer readable storage medium, including instructions stored thereon which when read and executed by one or more computers cause the one or more computers to perform a method comprising:

providing an on-device mobile application that operates in combination with an on-device speech-to-text engine to communicate with a data analytics system or environment for purposes of generating, modifying, or interacting with data visualizations, including:

listening for audio received as input from a user;

as the user provides input, detecting pauses for sentence consideration, and building an array of words associated with the sentence;

determining whether the sentence is capable of being processed without additional server data and if so performing the step locally; and

alternatively sending an input sentence to a server API, in order to determine an intent associated therewith for use by the server in returning relevant results.

12. The non-transitory computer readable storage medium of claim 11, wherein the data analytics system or environment is an analytics cloud environment.

13. The non-transitory computer readable storage medium of claim 11, wherein the data analytics system or environment includes a data analytics assistant provided by or for use with the data analytics environment in generating data visualizations associated with datasets.

14. The non-transitory computer readable storage medium of claim 11, wherein:

the data analytics system or environment receives as input a natural language expression, and the input natural language expression is associated with a context;

a search component finds a most relevant dataset, which is returned to a data visualization environment for rendering within a user interface.

15. The non-transitory computer readable storage medium of claim 11, wherein the data analytics system or environment and digital assistant system or environment are provided or communicate as part of a cloud environment.